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SAARLAND UNIVERSITY

Faculty of Mathematics and Computer Science  
Department of Computer Science  
MASTER THESIS

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# Evaluating User-Centric Aspects with Visual Explanations in Reinforcement Learning: A Game Case Study

submitted by

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# Abstract

In real-world applications, reinforcement learning RL has experienced growth across various fields, including autonomous driving, robotics, and games. However, its lack of transparency in the decision-making process hinders the user's understanding and trust. To address this challenge, explainable reinforcement learning XRL aims to make RL's decision-making process more transparent and understandable to users. Despite growing interest in XRL, there is still limited empirical evidence on how different explanation modalities work in interactive 3D environments and how they influence user-centric aspects such as trust and satisfaction during decision-making. This thesis examines the user's trust and satisfaction with visual explanations that are directly embedded into a 3D game environment, using *Ouch! Marble Stick* game as a case study. In this game, a player pulls out sticks from a bowl that holds marbles while causing the fewest marbles to fall down on the ground. Two explanation styles, such as colored marbles and shaking marbles animation, are implemented to indicate which features influenced the agent's chosen action. A controlled user study is conducted to evaluate the impact of these explanations on user trust in the agent and their satisfaction with the explanations. The findings indicate that colored marbles explanation style was preferred by the majority of the users, while shaking marbles animations were engaging but produced mixed reactions. Furthermore, both explanation conditions increased user satisfaction compared to no explanation.



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# Chapter 1

## INTRODUCTION

Artificial Intelligence (AI) has gained popularity across various domains, ranging from everyday applications such as recommendation and personalization on digital platforms (e.g., Amazon, Netflix) to high-stakes settings such as assisting experts in analyzing medical data and recommending treatment plans. In both cases, these systems improve their decisions by learning from past data and adapting their behavior according to user needs. This rapid progress in AI technologies is due to the recent advances in Machine Learning (ML) and Deep Learning (DL). They learn patterns from large amounts of data and make increasingly accurate predictions. As a result, AI has become capable of handling complex real-world tasks, including perception and decision-making in autonomous driving.

Reinforcement Learning (RL) is a branch of machine learning that focuses on making sequential decisions. An agent learns through repeated interaction with the environment by taking actions and receiving feedback. It is often combined with deep neural networks to build Deep Reinforcement Learning (DRL) agents that process high-dimensional inputs and learn effective behavior in challenging environments. Thus, DRL has attracted a lot of interest because it can achieve human-level performance in complex tasks that were previously difficult for traditional methods.

Despite these successes, AI, RL, and DRL often act as "black-box" systems. Their internal reasoning for taking a certain decision is represented by millions of parameters, which is difficult for stakeholders (e.g., users, developers, and domain experts) to understand. This lack of transparency and interpretability becomes a major concern when these AI agents have a significant impact on everyday life. As a result, there is a growing need for methods that can provide explanations of an agent's decision.

Explanations are primarily needed to improve the transparency and comprehensibility of the decisions taken by these agents. They can be used for several purposes, such as justifying why a decision was taken by the agent. With the help of explanations, developers can monitor and debug agent behavior by revealing faults in behavior and correcting them. Explanations can also be used to improve the agent's ability to perform a task. In addition, they can help discover new facts and gain knowledge about strategies used by the agent to accomplish the task [1, 2].

eXplainable Artificial Intelligence (XAI) aims to bridge the gap between high-performing black-box agents and the need for users to appropriately trust the agent’s decisions.

The focus of this thesis is to develop an XRL agent that can deliver visual explanations in a 3D environment. The emphasis is on using visual explanations to improve the interpretability of the agent’s behavior. This allows users to calibrate their trust in the XRL agent and be satisfied with its decisions.

## 1.1 Motivation

Today, RL and DRL have made significant progress, and one of the most famous applications is AlphaGo. AlphaGo learned to play the complex Chinese board game of Go and defeated the world champion, Lee Sedol. In the second match, AlphaGo made an unexpected but crucial move that had a very low probability of being played. This stunned many Go experts, who said that it was a move that no human would play, yet it was the key to the victory of Alpha Go [3]. This instance highlighted both the strength of AI systems and their opacity. They are capable of making highly effective decisions without providing clear reasoning that users can easily understand.

In the case of AlphaGo, an unexpected decision had no serious direct consequences. It mainly affected the outcome of the game. However, when an AI or RL system takes an unexpected decision in other high-stakes domains, it can have serious consequences. For instance, when a medical diagnosis system recommends an unexpected treatment for the disease without providing an explanation, users may not know whether the recommended treatment is justified, safe, and reliable [4]. Similarly, autonomous driving in transportation, security, and finance are other examples where incorrect decisions can lead to safety risks, financial losses, and ethical issues. This leads to distrust in AI systems, which is why they must be explainable to ensure user trust and safety.

Explanations help users understand how a model’s input relates to its output. For example, explanations can highlight the parts of the input that influenced a prediction or action. Many explanation techniques have therefore been developed and evaluated in 2D settings, such as game environments or image classification tasks. In these settings, explanations can be visualized as saliency maps, or heatmaps. They can also be represented using simple structures such as decision trees. However, many real-world RL applications, such as robotics, operate in dynamic 3D environments, where decisions depend on spatial relationships and interactions over time. In such a setting, users may benefit more from explanations when they are embedded directly into the 3D environment. This is because it aligns with how users naturally perceive and interact with spatial environments. Research on XRL in such 3D settings is still underexplored. This thesis aims to address this research gap by incorporating explanations into 3D space.

## 1.2 Research Questions

The primary objective of the Master’s Thesis is to develop an XRL agent that provides visual explanations for its decisions. By embedding these explanations directly into 3D space, we hope to provide users with more immersive and intuitive feedback, especially in dynamic, interactive



environments. This will be investigated through a case study featuring a game called *Ouch! Marble Stick*, where players take turns pulling out sticks from a bowl that holds marbles. The goal is to carefully pull out a stick while causing the fewest number of marbles to fall to the ground.

To evaluate whether these explanations help users understand and rely on the agent, we conduct a user study. It focuses on measuring user-centric aspects such as trust and satisfaction. Importantly, trust is measured at three points with each game scene: (1) before the stick is executed and pulled out from the bowl (2) after the stick is executed, and the outcome is observed, and (3) after the optimal stick is revealed, and its outcome is observed. The following research questions (RQs) guide this work:

- **RQ1:** How do different visual explanation styles influence the users' trust in the agent's recommendations?
- **RQ2:** How do different explanation conditions influence the direction of the users' trust across the three phases within a game scene (before execution, after execution, and after showing the optimal solution)?
- **RQ3:** How does the agent's error severity (number of marbles dropped) affect the direction of the users' trust after executing the agent's recommended stick?
- **RQ4:** How do different visual explanation styles impact the user satisfaction during gameplay?

### 1.3 Thesis Outline

The remainder of this thesis is structured as follows. To establish a solid foundation for this research, chapter 2 covers key aspects such as RL, XAI, XRL, explainability methods, and related work. Chapter 3 presents the conceptual framework and design rationale used for this thesis. Chapter 4 focuses on developing the game, training an XRL agent, generating explanations, and developing the user interface. Chapter 5 describes the user study methodology and presents the quantitative and qualitative results. Chapter 6 discusses these results and their implications. Finally, Chapter 7 concludes the thesis, summarizing the findings for each research question, discussing limitations, and outlining directions for future work.



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# Chapter 2

## BACKGROUND AND RELATED WORK

### 2.1 Background Concepts

#### 2.1.1 Reinforcement Learning

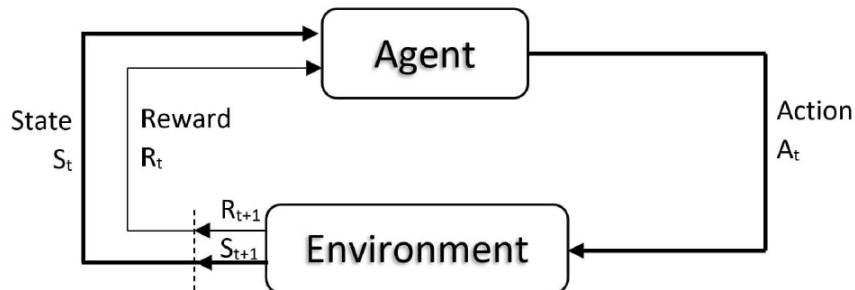


Figure 2.1: Agent-environment interaction cycle [2]

Unlike supervised learning, where models learn from labeled datasets, RL agents learn through interaction with their environment. It consists of agents, environment, states, actions, and rewards. Figure 2.1 depicts the agent-environment cycle where the agent is in a state  $S_t$ . It interacts with the environment and chooses one of multiple actions  $A_t$ , and the state changes to a new state  $S_{t+1}$ . The agent either receives a reward or penalty (negative reward) for the action taken, and the cycle repeats. The policy can help the agent to choose an optimal action in order to receive high rewards. Thus, the objective is to learn a policy that is either mapping from states to actions or a probability distribution over actions, that maximizes expected cumulative rewards [5]. Due to its ability to maximize the cumulative reward through trial-and-error, reinforcement learning is especially well-suited for tasks requiring long-term sequential decision-making. Therefore, reinforcement learning is applied across various domains including gaming, robotics,

autonomous driving, and many more.

### 2.1.2 Explainable AI and Explainable RL

With the increasing importance of understandability and transparency in AI models, XAI has attracted significant attention [6]. AI systems that are used in critical domains are not only expected to make accurate decisions but also to clearly explain those decisions. As defined by Arrieta et al. [7], “Given an audience, an explainable artificial intelligence is one that produces details or reasons to make its functioning clear or easy to understand.” This highlights the need for AI systems to provide meaningful insights into their decision-making processes and ensure that the reasoning behind the output is understandable to different audiences, whether they are developers, domain experts, or general non-expert end-users [6].

In order to comprehend an agent’s behavior in RL environments, Vouros [5] highlights a number of important factors.

- **Qualitative understanding** refers to how humans perceive the agent’s policies, objectives, preferences, and abilities. This enables them to deduce the agent’s reasoning behind specific decisions in dynamic contexts.
- The second factor he discusses is the **granularity and scale of explanations**. Granularity describes the level of detail in which an agent acts (e.g., particular actions or general behavioral patterns), while scale describes the temporal or spatial extent of the agent’s responses (e.g., across time or space).
- He draws attention to how **environmental features influence** the generation of explanations, with various feature subsets offering various degrees of insight into the agent’s choices.
- He emphasizes that explanations should take into account the **user’s beliefs, perceptions, and comprehension** of the inner workings of the system, and should be tailored to the needs of human operators who are going to use them in real-life rather than machine learning specialists. This will ensure that the responses are understandable and useful for real-world applications.

### 2.1.3 Explainability Methods

In figure 2.2, the explainability methods in AI and RL can be classified into two main categories: Intrinsic methods and Post-hoc methods [1]. Furthermore, based on the scope of the explanations, explainability methods can have a local or global scope of explanations [1]. Each category serves a distinct purpose in improving the interpretability and transparency of complex AI and RL models.

#### Intrinsic vs. Post-hoc methods

**Intrinsic methods** consist of models that are designed to be inherently interpretable. They are self-explanatory, as they provide clear insights into their decision-making process [2]. These

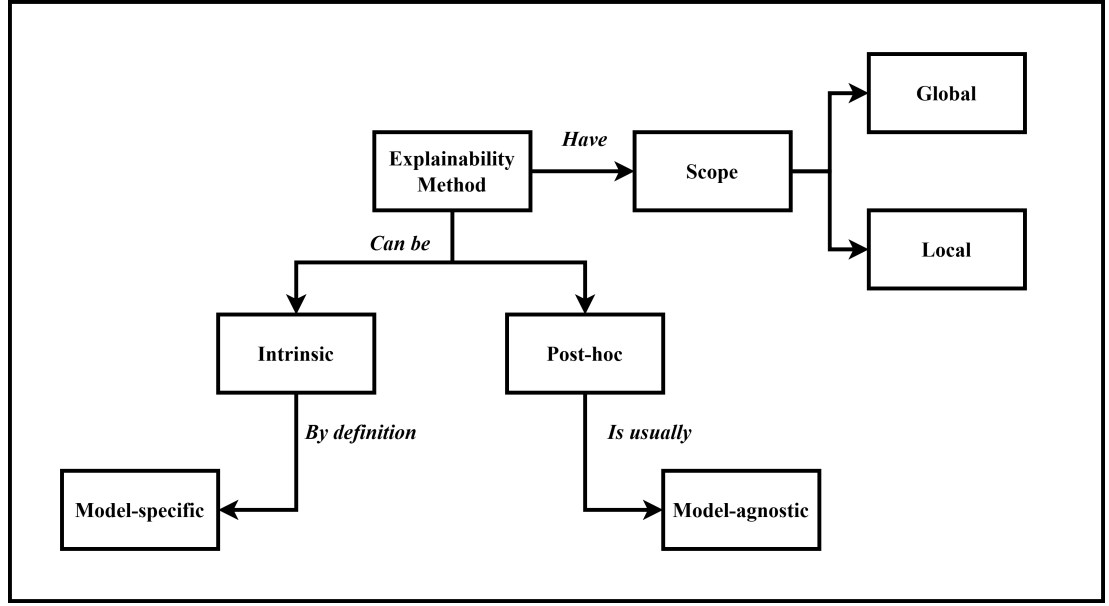


Figure 2.2: Classification of XAI methods [1]

insights are easily understandable to humans [2]. Examples of such models include decision trees, logistic regression, linear models, Bayesian models, or k-nearest neighbors [8]. A decision tree, for example, fully describes the steps and rules it uses to reach a decision, making it more transparent and understandable. Intrinsic explainability is therefore model-specific, implying that transparency is an inherent quality of the model.

Intrinsic methods offer built-in transparency because of their straightforward structure and decision-making process. However, it is more challenging to achieve similar interpretability in reinforcement learning. Standard Deep Reinforcement Learning (SDRL) algorithms, such as Deep Q-Learning (DQN), Proximal Policy Optimization (PPO), and Deep Deterministic Policy Gradient (DDPG), are not naturally interpretable. To address this issue, Heuillet et al. [9] summarized three methods to enhance RL explainability.

- The first method is **representation learning**, where algorithms learn low-dimensional, meaningful representation, and abstract features of state space that help to explain the model’s decision-making, particularly through State Representation Learning (SRL).
- The second method involves **simultaneously learning policies and explanations**, using techniques such as reward decomposition and minimal sufficient explanation.
- Finally, **hierarchical goal-based method** in RL divides main tasks into sub-goals, where low-level agents learn the optimal sub-goals and allow the high-level agent to form an interpretable representation of the environment, often supported by Hindsight Experience Replay (HER) for maximizing learning from past actions.

**Post-hoc explainability methods**, on the other hand, are applied to models that are not inherently interpretable, such as deep neural networks. These models are referred to as highly complex uninterpretable black-box models. They use a separate set of techniques to provide

explanations for their decisions without modifying their inner network [1]. Post-hoc methods are also referred to as model-agnostic by definition as they can be applied to any complex model after it has been trained, such as feature importance, sensitivity analysis, Local-Interpretable Model-agnostic Explanations (LIME) [10], SHapely Additive exPlanations (SHAP) [11] or Layer-wise Relevance Propagation (LRP) [12].

In the case of RL, Heuillet et al. [9] summarized two methods: **explanation through saliency maps** and **explanation through interaction data**.



Figure 2.3: Example of saliency maps [13]

- Saliency maps, also called heat maps, are visual tools used to highlight the most significant pixels or areas in an image or video that influence the decision of an AI model [13, 14, 9]. A simple example of the saliency maps is shown in figure 2.3, where the important feature of the dog is highlighted for image classification. In reinforcement learning, saliency maps can be useful for identifying which regions of the current game frame are most important for the agent's action selection. They help users understand what the agent attends to during the decision-making process. Since saliency maps provide an intuitive visual clue, they can guide users to notice specific relevant parts of an input, and they are often understandable even for non-experts [14].
- In contrast, explanation via interaction data entails examining data gathered from the agent's interactions with its surroundings. This approach provides more in-depth insights into the agent's behavior.

The key difference between intrinsic and post-hoc methods lies in the fact that intrinsic methods are transparent from the start, whereas post-hoc methods use additional techniques to interpret the decisions of complex models.

### Local vs. Global explanations

**Local explanations** focus on providing explanations for individual predictions or instances [1]. They aim to explain why a model made a specific decision for a given input. For instance, in the case of reinforcement learning, a local explanation could help to understand the reason for the agent taking a particular action in a certain state. These explanations are essential for increasing the user trust in the model outcomes, as they allow users to inspect decisions on a case-by-case basis [1]. Techniques such as LRP, LIME, and SHAP are common post-hoc local methods used to explain individual predictions in black-box models.

**Global explanations**, on the other hand, aim to provide insights into the model as a whole. These methods explain the overall decision-making process of the model [1]. They offer a broader understanding of model behavior across all inputs and outputs. For example, a global explanation in reinforcement learning could reveal general strategies that an agent has learned over time [4]. Global explanations are valuable for understanding the underlying mechanism of the model and ensuring that the decision-making process aligns with the intended goals [1]. However, because they summarize the model's behavior across all possible inputs, global explanations can sometimes be less faithful to individual predictions compared to local explanations [1].

### 2.1.4 Shapely Additive Explanations

There are many existing explanation methods such as LIME, DeepLIFT, LRP that have been proposed to interpret individual predictions of complex models. However, these methods are based on different assumptions and design choices. Therefore, it is often unclear how they relate to one another. This also makes it difficult to compare them directly to know which method provides explanations with desirable guarantees, and understand under which conditions one method is preferred over another [11]. To address this key problem, Lundberg et al. [11] proposed SHAP a unified framework for local feature attributions. It is a game-theoretic framework based on Shapely values from cooperative game theory. It is widely used for explaining individual predictions of complex machine learning models.

SHAP is based on the idea that an explanation for a prediction can itself be viewed as an explanation model  $g$ . It is defined as a simple, interpretable approximations of the original prediction model  $f$  for a specific input  $x$ . SHAP focuses on additive feature attribution method, where the explanation model is a linear function of binary input features, indicating whether each feature is "present" or "absent". The explanation takes the form:

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i$$

where,  $z' \in \{0, 1\}^M$ , and  $z'$  is a simplified binary representation which indicates whether a feature is "present".  $M$  is the number of simplified input features.  $\phi_i$  represents the contribution value of input feature  $i$  and  $\phi_0$  is the baseline prediction or the expected outcome when no features are known [11].

Intuitively, each feature is treated like a "player" in a game, and the model prediction is the "payout." SHAP assigns a contribution value, also called as SHAP value, to each input feature for a local instance [11]. SHAP values answer the question: How much does knowing feature  $i$  contribute to the expected prediction, when averaged over all possible combinations of features? The SHAP value of a feature corresponds to its average marginal contribution to the payout across all possible combinations of features.

Figure 2.4 illustrates a local SHAP explanation for a single prediction. The model predicts the mortality risk score and TreeExplainer, a SHAP-based method for tree models, is used to explain the prediction. It decomposes the prediction as the sum of the baseline prediction and additive feature contributions. Each feature received a SHAP value that indicates how much it

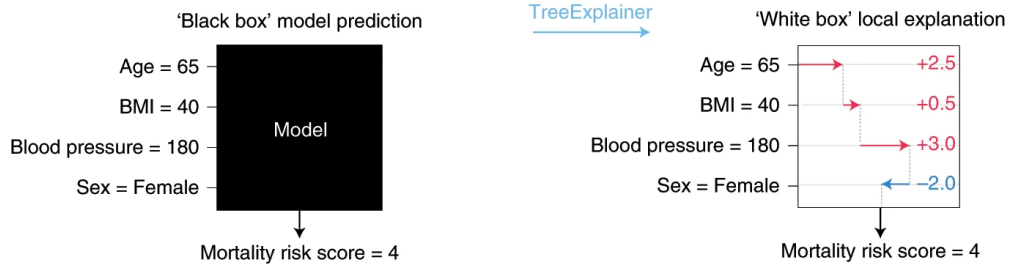


Figure 2.4: Example of SHAP [15]

increased or decreased the final model prediction [11, 15].

$$f(x) = \mathbb{E}[f(x)] + \sum_{i=1}^M \phi_i$$

where,  $x$  is the input observation,  $f(x)$  is the predicted value of the model, given input  $x$ ,  $\mathbb{E}[f(x)]$  is the expected value of the target variable and  $\phi$  is the contribution value of input feature  $i$ .

SHAP also satisfies three properties [11]:

- **Local accuracy:** the explanation should match the model output for the instance that is being explained. That means, the sum of the contribution values for all features should reconstruct model prediction.
- **Missingness:** if a feature is absent in a simplified representation, it should receive zero attribution.
- **Consistency:** if a model changes such that a feature's contribution increases or stays the same across all context, that feature's attribution should not decrease.

## 2.2 Prior Research in Explainability for AI and RL Systems

This section discusses major research in explainability for AI and RL systems, emphasizing how XAI and XRL frameworks were implemented and evaluated through user studies. The reviewed literature is divided into three primary categories. First, we delve into XAI frameworks, including the criteria and design concepts proposed by researchers for creating explainable systems. Next, we look at XRL applications, focusing on how explainability approaches were utilized in games like Frogger, Atari, and Pacman to make the behavior of RL agents more understandable. Finally, we investigate user-centric elements of XAI, namely how user studies assess crucial criteria such as trust, and satisfaction when dealing with explainable systems.



### 2.2.1 Frameworks of XAI

In this section, we explore frameworks that address how explanations can be effectively integrated into AI systems, ensuring that users can access relevant and interpretable information.

Xu et al. [6] presented a set of eight guidelines for designing effective XAI systems supporting user-centered explanations. These guidelines are structured around three fundamental questions of *when*, *what*, and *how* explanations should be provided. When developing an XAI framework and generating explanations for the decisions taken by AI, developers should consider these three main questions [6]. First, when explanations should be presented. This involves analyzing if every action needs an explanation and determining the timing for showing the explanation to users. Under this category, Xu et al. [6] proposed the following guidelines:

- **G1:** Make explanations always accessible to the users
- **G2:** Trigger explanations upon users' request.

Second question is What type of explanation content to provide? What should be the depth of the explanation content? It is important to select an appropriate type of explanation (e.g., Why-Why not, Certainty, etc.) and the length of explanation to meet user needs without overwhelming them. For this, Xu et al. [6] proposed following guidelines:

- **G3:** Show personalized explanation content by considering three factors: system goal, user goal, and user profile.
- **G4:** Display concise explanations answering "*why*"
- **G5:** Always provide opportunity to explore detailed explanations upon request.

Last question is How to deliver the content using appropriate modality? Depending on the cognitive load of the user, it is crucial to select an appropriate modality for the delivery of the explanations in interactive and immersive environments. The authors proposed the following guidelines:

- **G6:** Adopt same modality as the AI output
- **G7:** Use text as primary explanation format and employ graphics if they are easy to understand
- **G8:** Use implicit visual patterns when explanation content can be embedded in the environment. Otherwise, use explicit visual patterns.

In a similar effort to make AI decisions interpretable, Wang et al. [16] developed a conceptual user-centric XAI framework that linked the human reasoning process with XAI techniques. Later, they implemented it in a real-world setting as a medical diagnostic tool for early decision aid to diagnose patients in intensive care units. They used XAI approaches like SHAP for feature attribution to deliver explanations in the form of visual aids such as 2D plots and visualizations in Tableau as shown in figure 2.5, illustrating that a similar approach can be applied in other domains.

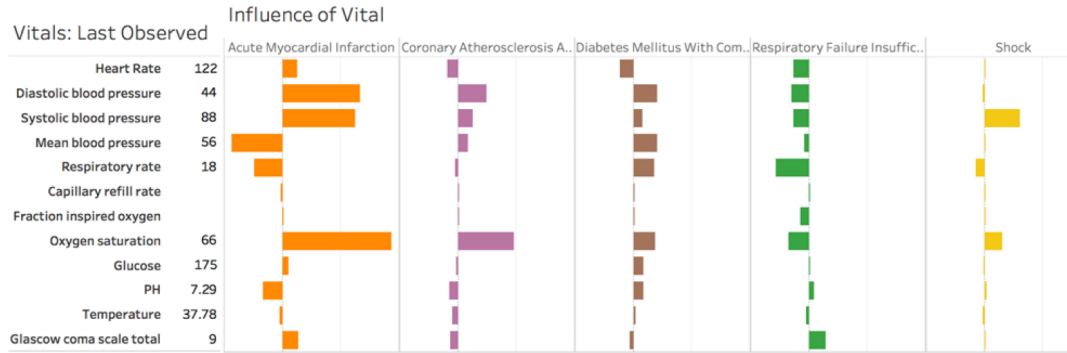


Figure 2.5: Screenshot of the AI-driven medical diagnosis tool with explanation sketches showing a patient with high predicted risk of acute myocardial infarction (AMI), heart disease, diabetes with complications, shock, etc [16]

### 2.2.2 Application of XAI and XRL in Games and Visual Tasks

Saliency maps have been widely utilized as a local explanation technique to understand the behavior of AI systems in a variety of tasks. In settings with static 2D inputs, such as image classification, saliency maps highlight the salient regions of an image that influenced the model's decision. For example, Alqaraawi et al. [14] used saliency maps to provide local explanations by visualizing the parts of an input image that were most relevant to the model's decision. Beyond classification, saliency maps have also been applied in complex tasks performed by RL agents, such as explaining the decisions made by the agent while playing 2D video games [17, 4, 18]. In this context, they highlight the areas of the game environment that impacted the agent's behavior.

Greydanus et al. [17] used saliency maps exclusively to investigate how agents learn and execute policies and track the development of the agent's strategy throughout the training process. Figure 2.6 shows how the attention to important areas of the environment changes during the learning process [17]. During the early stages of training, the saliency maps are unfocused. As the training progresses, the agent's focus becomes more concentrated on relevant regions of the game and reflects the emergence of effective strategies. In this context, it learns the significance of a tunneling strategy [17]. This method helped the authors to identify reasons behind the low-quality decisions and thereby suggest ways for improvement [17]. It is also crucial to determine whether the agent is overfitting to reward functions by taking decisions for the wrong reasons.

Alqaraawi et al. [14] and Greydanus et al. [17] both evaluated the saliency maps for local explanations and concluded that saliency maps indeed helped users pay attention to a specific area of interest that impacted the agent's decisions. They can provide significant and useful insights for understanding an agent's behavior. Moreover, the authors also identified the limitation of exclusively using the saliency maps and suggested incorporating an additional explanation technique.

To address the limitations of using saliency maps alone, several studies have explored combining them with other explanation techniques. Alqaraawi et al. [14] combined saliency maps with classification scores to provide a detailed explanation of agent behavior. Similarly, Anderson et al. [18] paired saliency maps with the reward decomposition technique. It broke down the



Figure 2.6: Visualizing learning process of the agent. Frames are chosen from the game played by a fully-trained agent. Leftmost agent is untrained, rightmost agent is fully trained. Each column is separated by ten million frames of training. White arrow denotes the velocity of the ball [17]

reward function into six different types of rewards to enhance the user’s understanding of the agent’s decision-making process in real-time strategy games. Huber et al. [4] explored how integrating saliency maps with strategy summarization, a global explanation method, can provide a more comprehensive understanding of agent behavior. Strategy summarization extracts the key states from agent simulation and generates a summary of the overall behavior of the agent throughout the game. This combination not only provides insights for specific decisions but also gives a high-level overview of the agent’s goal. This suggests that a holistic approach, integrating both local and global explanations, offers a more complete picture of the agent’s decision-making process.

In addition to saliency maps, researchers have also emphasized the role of dynamic visualization and animation for explaining the AI behavior in time-evolving tasks. For example, Guan [19] discusses how animations and interactive visualization can help make AI-driven cybersecurity analysis more understandable. The author used animations to visualize the changes in network traffic over time, highlight anomalies as they emerge, and animate intrusion paths to show how attacks progress through a system [19]. These animations helped to perceive the threat faster and make informed decision.

Other researchers such as Coppens et al. [20] and Ehsan et al. [21] have proposed alternative approaches to make reinforcement learning agents more interpretable through different forms of explanation. Coppens et al. [20] introduced Soft Decision Tree (SDT) distillation to understand policies learned by an RL agent playing the Mario AI benchmark. By transforming the output of the deep policy network into soft decision trees, it gave a hierarchical perspective of the DRL agent controlling Mario and its decision-making process, allowing users to track how input features impact action selection. On the other hand, Ehsan et al. [21] focused on automated rationale generation. They aimed to translate an RL agent’s internal decision-making process into natural language explanations by training a model on a human-explained dataset. The agent is playing the Frogger game, where the player is controlling a frog to move from the grassy strip at the bottom to the lilypads at the top. The player has to first cross the road without getting hit by cars, and then cross the river by jumping on logs. The rationale generator translates the current state of the environment and the action taken by the agent into natural language in order to produce human-like rationales for their actions [21].

To summarize, all the approaches mentioned above highlight the importance of exploring diverse methods to make RL agents more transparent and comprehensible to the users.

### 2.2.3 User-Centric Aspects of XAI

Various techniques have been developed to improve the interpretability and transparency of AI systems such as saliency maps, natural language, decision trees, and many more. However, it is also essential to evaluate the effectiveness of these explanations from a user-centric standpoint. Thus, in this section, we discuss the concept of user trust, and user satisfaction. These aspects help us determine whether the explanations promote clarity, positive user experience, and confidence.

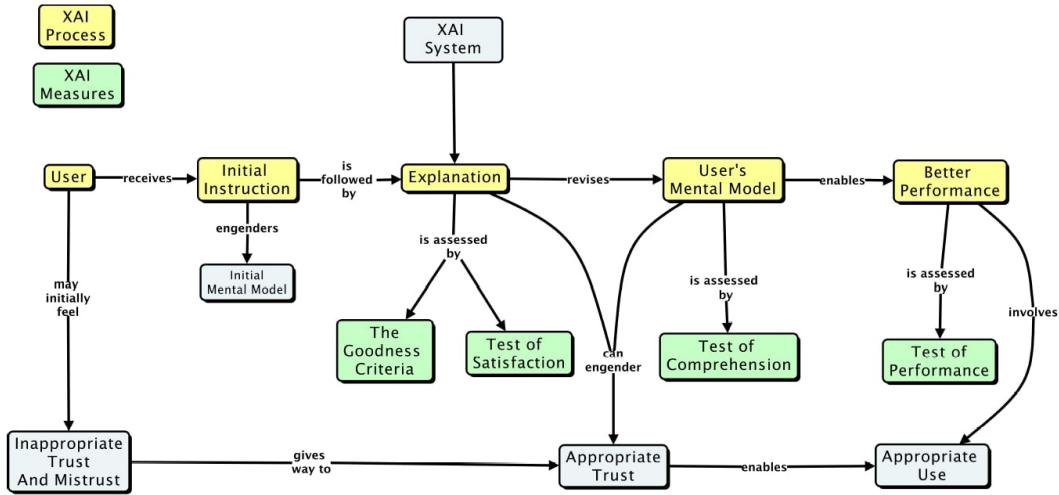


Figure 2.7: Conceptual model of explanation process in XAI [22]

Figure 2.7 depicts a conceptual model of the explanation process in the XAI environment, focusing on how people engage with AI systems and develop confidence. It starts with the user, who may first have faith or mistrust in the system. After receiving initial instructions, the user creates an initial mental picture of how the system operates. After interacting with the system, users are given an explanation by the XAI system, which is evaluated by using two metrics: the goodness criteria, which evaluates explanation quality, and the test of satisfaction, which assesses how happy and satisfied the user is with the explanation. This explanation serves to develop the user's mental model, which is subsequently tested by a test of comprehension, which measures user understanding, and a test of performance, measuring how the explanation improves their use of the system. As the user has a better grasp of the system, their faith in the XAI systems develops. If the user initially mistrusted the system, the explanation and this procedure seek to convert this mistrust into acceptable confidence. Finally, this improves system performance and proper usage, guaranteeing that the user trusts the AI in the right settings and uses it successfully.

#### User Satisfaction

Even if explanations may be technically good and may lead to better decision-making, they may still fail. Users may find them confusing, distracting, or unhelpful. Hence, Hoffman et al. [22] defined explanation satisfaction as the degree to which users feel they understand the AI system based on the explanations provided. Satisfaction is defined as a “contextualized, a

posterior judgment” and is shaped by users’ interaction with the explanation after it is delivered [22]. Key attributes of explanation satisfaction include understandability, feeling of satisfaction, sufficiency of detail, completeness, usefulness, accuracy, and trustworthiness [22]. These factors are typically measured through psychometric measures, like Likert scales [22].

### User Trust

User trust is an important factor when it comes to the adoption of XAI systems. For users to effectively use AI and RL systems in real-world scenarios, they must trust the system’s decisions. Therefore, evaluating user trust through user studies becomes crucial.

Human-computer trust is defined as “the extent to which a user is confident in, and willing to act on the basis of, the recommendations, actions, and decisions of an artificially intelligent decision aid” [23]. According to Yang et al., trust in AI has two main components: confidence in the system’s decisions and the willingness to follow its recommendations [24]. Figure 2.8 shows the formulation of objectively evaluating trust through observable user behavior of whether to follow or reject system’s recommendation and different outcome conditions, that is, whether the system’s recommendation is correct or incorrect [24, 25].

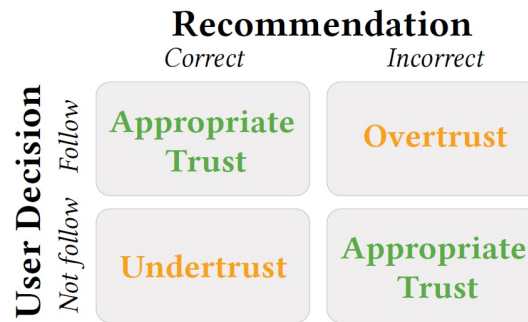


Figure 2.8: Cases that are classified as appropriate trust, overtrust and undertrust [24]

It is crucial for users to have appropriate trust in the AI systems, which is different from overtrust and undertrust [24, 25]. Yang et al. [24] trained an image classification model and used XAI techniques to explain the system’s decisions for the user study to evaluate the trust in the model. Participants were asked to classify images and assess whether they followed the model’s explanations. Based on the user’s follow decision and the model’s ground truth, the authors defined all the trust labels: If a participant followed the classifier’s recommendation when it was correct, and if participants not followed the classifier’s recommendation when it was wrong, this was considered appropriate trust [24, 25]. However, if the participant followed the classifier even when it was wrong, this indicated overtrust [24, 25]. Conversely, undertrust occurred when the user chose not to follow the classifier’s recommendation, even when the model was accurate [24, 25].

Unlike Yang et al. [24], our approach doesn’t have the ground truth to compare the participant’s response with the action taken by the agent and whether the action is the right or wrong decision. Huber et al. [4] devised another approach to evaluate trust by asking the participants to perform the agent comparison task. They did not directly measure the user trust, but they asked them to compare the performance of the 2 different agents playing the Pac-Man game

and chose one agent that they would trust to play on their behalf [4].

Yang et al. [24] also measured subjective trust using a linear trust meter. The trust meter is a slider, and its value ranged from completely distrust (-100) to completely trust (+100). Participants were encouraged to adjust trust whenever their trust changed. This captured how trust evolves during interaction.

## 2.3 Research Gap

Many researchers have demonstrated the importance of visual explanations for making AI and RL systems more transparent and interpretable. In medical diagnosis systems, researchers have used visual aids such as plots and dashboards to show a model’s reasoning to medical practitioners [16]. Researchers have also used saliency maps to provide local explanations in gaming environments [4, 17, 18] and in image classification tasks [14]. Furthermore, Coppens et al. [20] investigated soft decision trees to understand policies learned by an RL agent in a gaming environment. However, these approaches are implemented in 2D space, where explanations are shown as overlays on images, or on game screens, or as external plots and decision trees.

In addition, many interactive RL settings are inherently dynamic and unfold over time. While prior work in games relies on static visualizations such as saliency maps overlays, or decision trees, Guan [19] emphasized the importance of animation-driven visualization in the cybersecurity domain to make AI outputs more interpretable in dynamic scenarios. This suggests that animation can complement static visual explanations in time-evolving environments. However, animation-based explanation styles are still underexplored in explainable reinforcement learning, particularly in 3D settings.

Alternative explanation forms such as textual explanations can be useful in some domains. However, they are less suitable for an interactive 3D environment because reading text can disrupt immersion and break the flow of the game. It can also increase the cognitive load. For interactive environments, visual cues, such as highlighting certain areas of interest or applying animations, blend smoothly into the 3D world. They ensure immersion and promote quicker comprehension and engagement. Therefore, this thesis focuses on visual explanation styles that seamlessly integrate into a 3D environment.

To address these gaps, this thesis embeds local visual explanations directly into a dynamic 3D interactive game environment. Instead of using external plots or text-heavy explanations, this thesis provides visual explanations through color-based highlighting and animation-based cues. These explanation styles preserve immersion and communicate feature relevance within the game. Finally, this thesis systematically evaluates user-centric aspects, such as trust calibration and satisfaction with these explanations styles, through a user study. Together, these contributions extend prior 2D-focused work toward more immersive and user-centered explainable reinforcement learning in 3D settings.

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# Chapter 3

## CONCEPTUAL FRAMEWORK AND DESIGN RATIONALE

After understanding the background concepts, reviewing selected related work and identifying the research gap in the field of XAI, this chapter presents the conceptual framework which was used to conduct the user study. It covers all the design decisions taken in this thesis.

This thesis explores explainability in reinforcement learning from a user-centric standpoint. It focuses on the following aspects:

- Developing an explainable RL agent for a physics-based 3D game environment and integrating visual explanations directly inside the 3D space.
- Designing and implementing multiple visual explanation styles, such as color-based highlighting and animation-based cues that indicate which game elements influenced the agent’s recommended action.
- Conducting a controlled user study to evaluate how these visual explanations impact users’ trust in the XRL agent and their satisfaction with the explanations provided by the agent during interaction.

Today, reinforcement learning agents are increasingly used to accomplish complex tasks in 3D space, such as gaming, robotics and simulations. However, their action depends on the complex spatial relationships, and it is often difficult for end users to understand why the agent took a particular action. Thus, it is crucial to make these agents explainable, as the explanations can help users to understand and foster trust for effective human-AI collaboration.

To address this challenge, we investigated explainable reinforcement learning in the context of a 3D game environment based on the *Ouch! Marble Stick Game*, where pulling out a selected stick from the bowl can cause the marbles to fall down due to gravity. The game was appropriate for this work because of two reasons. First, it offers physically an intuitive 3D space setting where

an agent’s actions have observable consequences. Second, it is an interactive game and well suited for user study to evaluate how explanations help users to trust the agent.

Since the game provides a 3D spatial and dynamic setting, players mainly focus on visual cues such as positions of objects or movements of marbles in the game. Visual explanations are inherently compatible because users can easily relate with the agent’s reasoning to objects inside the game. Thus, we focus on providing explanations in visual form.

Furthermore, we focus on providing local explanations that explain the individual decision taken by the agent for a given game state. As the game environment changes continuously, users are more likely to seek an explanation for a specific in-game decision rather than a global one, summarizing the overall agent behavior. Local explanations can provide clarity on every moment decision-making processes, which is essential in interactive environments such as games.

We discussed various categories of explainability methods in section 2.1.3. We focus on using post-hoc methods to generate explanations for the reinforcement learning agent. This choice is motivated by the following reasons:

- **Flexibility Across Models:** Post-hoc methods can be applied to any complex model, without needing to modify the internal model structure. This flexibility allows us to work with complex DRL algorithms and achieve optimal performance. On the contrary, intrinsic models for reinforcement learning require learning states, policies, and explanations. This can restrict the choice of architectures and limit scalability.
- **Performance-Preserving:** Post-hoc methods do not interfere with the agent’s learning process. These methods are applied on top of the trained model. Contrary to that, intrinsic methods may compromise the performance of the agent as it needs to learn explanations during training.
- **Adaptability to Different Types of Explanations:** Post-hoc methods like LIME, SHAP, and LRP provide explanations at different levels of granularity. These post-hoc explanations can include feature importance, specific input-output explanations, or relevance propagation. Once the agent is trained, we can adapt to different types of explanations as per the user’s needs or the situation. On the contrary, intrinsic methods tend to be tied to the specific explanation framework used during training, which may limit their adaptability in real-time decision-making scenarios.

Among post-hoc explanation methods, SHAP is a well-established, reliable and widely adopted XAI method. This thesis uses SHAP to generate explanations for the agent’s decision because it offers local, additive explanations that quantify the contribution value of each input feature to a specific decision. These contribution values are consistent and comparable. These can be mapped to visual cues such as coloring or shaking marbles to provide intuitive visual explanation in a 3D space.

In order to build an effective XAI system, we considered the eight design guidelines proposed by Xu et al. [6]. These guidelines outline best practices for making explanations accessible, non-intrusive and aligned with user needs. In this work, we fully followed five guidelines (G1, G2, G4, G6, G8), partially followed two guidelines (G3, G5) and not followed one guideline (G7) intentionally due to the interactive and spatial nature of the game in the user study. A



detailed discussion of how these guidelines were integrated in the user interface is provided in section 4.4.

After defining how explanations should be generated, and presented, the next crucial design choice is which user-centric aspects should be evaluated and how to measure them. This thesis focuses on evaluating user trust and user satisfaction with the 3D visual explanations because appropriate trust in the agent and user satisfaction with the agent’s explanation are essential prerequisites for adopting an effective XAI system.

In order to measure trust, we adopted two approaches: behavioral trust calibration and self-reported trust value in agent via trust meters.

- Behavioral trust calibration captured whether participants relied on the agent appropriately [24]. We developed a trust classification logic that classifies each user-agent interaction into *appropriate trust*, *overtrust*, *undertrust*. This classification logic is objectively based on (1) whether participants followed the agent’s decision, and (2) the outcome of the stick choice.
- We displayed trust meter after 3 distinct phases. This captured how trust evolved during the entire gameplay [24].

In conclusion, this chapter established the conceptual foundation and design rationale underlying the proposed approach of explainable reinforcement learning in a 3D environment. It provided justification for the choice of visual, local, post-hoc explanations and evaluating user trust and satisfaction. Building on these decisions, the next chapter describes the implementation of the game environment, training of the agent, explanation generation pipeline and the user interface.



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## Chapter 4

# IMPLEMENTATION

This chapter describes the implementation of the proposed approach introduced in the previous chapter of this thesis. In the beginning, it introduces the *Ouch! Marble Stick Game*. It explains how it was adopted and developed into a digital 3D environment in section 4.1. Section 4.2 explains the training process of the RL agent for the game. Finally, the chapter describes the XAI method and presents two visual explanation styles integrated in the game.

### 4.1 Ouch! Marble Stick Game

The *Ouch! Marble Stick Game*, also known as *Junior Ouch!* or *Big Ouch!*, is a tabletop skill, strategy, and physics-based multiplayer game (2-4 players) for children. It develops fine-motor skills and hand-eye coordination. It aims to increase concentration and attention span. Players take turns removing one stick per round. The goal is to pull out one stick per turn using a steady hand while causing the fewest marbles to fall on the base. The core idea of the game is that the players must maintain balance between marbles and other sticks while pulling out the chosen sticks from a structure. The game ends when all the sticks are removed or all the marbles have fallen. The player who causes the fewest marbles to drop wins the game. Figures 4.1a illustrates the original game.

#### 4.1.1 Game Components and Setup

The standard *Ouch!* game consists of a base tray, a transparent tower (a cylindrical hollow tube), a transparent perforated bowl, sticks, and marbles, as shown in figure 4.1b. The base tray is first placed on a stable platform, and the tower is mounted on the base. The bowl is then placed on top of the tower. There are a total of 24 sticks in four different colors (red, blue, green, and yellow). The sticks are inserted through opposing holes in the transparent bowl at different heights and angles, creating a crisscross mesh or web-like structure. About 28 black marbles and 2 white marbles are gently placed on the stick mesh until they come to rest. The difficulty



Figure 4.1: Ouch! Marble Stick Game and its components

level of the game can be adjusted by modifying the configuration of the sticks by inserting the sticks at different angles and at different heights.

#### 4.1.2 Game Development in Unity

The digital version of the *Ouch!* game was designed and developed using the Unity Engine (editor version 2022.3.55f1). The transparent bowl and the cylindrical tower were first created in Blender (software version 4.1). These objects were later exported as FBX objects and imported into the Unity Engine. Additional objects such as sticks, marbles and the base tray were constructed using primitive shapes in Unity Engine. A Rigidbody component was attached to all GameObjects to enable physics-based interactions such as gravity and collisions. A Capsule Collider was attached to each stick, and a Sphere Collider was attached to the tower and each marble.

Figure 4.2 shows the 3D digital version of the *Ouch!* game developed to evaluate user-centric aspects with 3D visual explanations. There are a total of 11 sticks and 17 marbles. The sticks and marbles were placed procedurally to form a stable structure inside the bowl. At the start of the scene, a seed was selected from 1 to 10000 and used to initialize the random generator, making placement of the sticks and marbles deterministic for later reconstruction. The sticks were first inserted into the bowl from different directions, angles, and heights to form a web-like crisscrossed structure of sticks inside the bowl that could support marbles. During placement, each newly inserted stick was validated against the already placed sticks so that the structure was physically plausible and unrealistic overlaps were avoided.

Once all the sticks were placed, marbles were spawned one by one above the bowl, allowing them to settle naturally under gravity onto the web-like stick structure. If a marble passed through the gaps between sticks and fell through the bowl opening into the tower or failed to settle reliably within a short time window, it was discarded and re-spawned. This was repeated until a stable placement was achieved. To avoid endless retries, placement attempts were bounded and time-limited.

The digital version of the game closely replicated the structural design and physical behavior

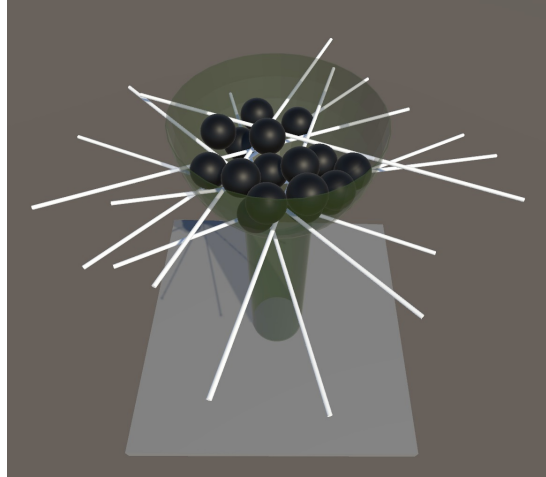


Figure 4.2: Digital version of the Ouch! Marble Stick Game developed in Unity

of the original tabletop game. When a stick was pulled out, the marbles responded realistically under the influence of gravity. To detect and count marble drops, a spherical trigger collider was placed inside the tower at approximately mid-height. Whenever a marble fell through the bowl opening into the tower and hit the trigger collider, the drop counter was incremented.

Although the original game is designed for multiple players, the digital version was intentionally designed as a single-player interactive game. It allowed for a controlled experimental setting for the user study because neither human vs. human nor human vs. agent competition would produce consistent, comparable results across participants. In addition, all sticks were colored white, and all marbles were black. This color scheme was chosen intentionally to avoid visual confusion and to maintain clear visual context when overlaying color-based visual explanations.

## 4.2 Agent Design and Training

Predicting which sticks are essential to maintain the stability of marbles is difficult because pulling out an incorrect stick could trigger a cascade of falling marbles. To capture this complexity, an RL agent was trained for the digital version of the game to identify which stick should be pulled out in a given game state. The agent was implemented using the Unity ML-Agents Toolkit [26]. It is an open-source framework that allows interaction with simulated environments created in the Unity Editor via a Python API for training and evaluation. It includes an ML-Agents SDK with the core C# components. These components are needed to define agents, observations, actions, rewards, and a training pipeline in Unity. It converts the Unity scene into a learning environment for training agents using various algorithms.

The toolkit also includes state-of-the-art RL algorithms such as PPO [27]. It was employed as the learning algorithm to train the agent for playing the game. It is an online, on-policy, policy-gradient RL method. It is an actor-critic method that alternates between collecting data by interacting with the environment and then performing several epochs of optimization on a "surrogate" objective function using stochastic gradient ascent [27]. It provides stability and

efficiency in the learning [27].

At the start of each training episode, the environment was initialized by inserting 11 sticks into the bowl and dropping 17 marbles inside the bowl and resting onto the stick web-like structure. This initial placement of sticks and marbles was random across the training process, using different random seed values to enhance generalization and prevent overfitting to specific configurations. A pseudo-random seed was set at the beginning of each episode before placing the sticks and marbles. This made each generated configuration reproducible when the same seed is reused. Moreover, the seeds used for training the agent and for inference (game scenes) were different.

**Observation (state) vector:** Once the environment was initialized, the agent observed a fixed-length vector of the current state of the environment. It encodes the spatial configuration of sticks and marbles. Let the observation at time step  $t$  be  $o_t \in \mathbb{R}^{128}$ , constructed as the concatenation of (i) normalized marble positions  $(b^x, b^y, b^z)$ , (ii) normalized stick positions  $(s^x, s^y, s^z)$ , and (iii) stick rotations  $(q^x, q^y, q^z, q^w)$ .

$$o_t = \left[ \underbrace{b_0^x, b_0^y, b_0^z, \dots, b_{16}^x, b_{16}^y, b_{16}^z}_{17 \times 3}, \underbrace{s_0^x, s_0^y, s_0^z, q_0^x, q_0^y, q_0^z, q_0^w, \dots, s_{10}^x, s_{10}^y, s_{10}^z, q_{10}^x, q_{10}^y, q_{10}^z, q_{10}^w}_{11 \times (3+4)} \right]$$

If a marble or stick is unavailable, its corresponding entries are set to zero. This state representation enabled the agent to reason about the spatial arrangement of sticks and marbles and anticipate the effect of pulling out a particular stick on marble stability.

**Action space and handling invalid actions:** Once the observation vector was given to the agent, the agent used it at each decision step to select an action. The agent operated in a discrete action space with 11 actions, where each action corresponded to pulling out a stick from the bowl. Once a stick was pulled out, the action corresponding to that stick was masked out at the next decision time. This prevents the agent from selecting already pulled-out sticks and ensures that it only selects feasible actions.

**Reward function:** After each stick was pulled out, the environment observed the number of marbles that fell through the bowl opening into the tower. The agent received either a positive reward of +3 points if no marbles fell down or a penalty of -1 point for every marble dropped after pulling out the stick. This reward function encouraged the agent to select a stable stick that does not cause marbles to fall down and discouraged the agent when marbles fell down.

**Episode length:** Each episode terminated once the agent had pulled out 6 sticks. Because after the initial placement of sticks and marbles, there are often multiple sticks that are safe to pull out without causing marbles to fall down. Furthermore, pulling out one stick changes the stability of the entire structure. That means a stick that was safe early in the episode may become unsafe later, and vice versa. Thus, this created a consistent learning space and trained the agent to maintain the stability across multiple stick removals rather than just optimizing the agent only for the first single safe stick removal. This process is repeated for many episodes, and the agent performance improves over time.

**Policy network architecture and PPO hyperparameters:** Since the environment provided a fixed-length vector observation, the agent used a fully connected multilayer perceptron (MLP) policy/value network. The agent was trained with Proximal Policy Optimization (PPO). We used standard ML-Agents PPO as a baseline (shown in Table 4.1), which provided a good starting point for training in continuous physics-based environments.

| Setting                               | Value                                  |
|---------------------------------------|----------------------------------------|
| <i>Network settings</i>               |                                        |
| Hidden layers (num_layers)            | 2                                      |
| Hidden units per layer (hidden_units) | 128                                    |
| Observation normalization (normalize) | true                                   |
| <i>PPO hyperparameters</i>            |                                        |
| Batch size (batch_size)               | 64                                     |
| Buffer size (buffer_size)             | 2048                                   |
| Learning rate (learning_rate)         | $3 \times 10^{-4}$ (linear schedule)   |
| Entropy coefficient (beta)            | $5 \times 10^{-4}$ (constant schedule) |
| PPO clip parameter (epsilon)          | 0.2 (linear schedule)                  |
| GAE parameter (lambda)                | 0.95                                   |
| Epochs per update (num_epoch)         | 3                                      |
| Discount factor (gamma)               | 0.99                                   |
| Time horizon (time_horizon)           | 64                                     |
| <i>Training budget and logging</i>    |                                        |
| Maximum training steps (max_steps)    | 200,000                                |
| Summary frequency (summary_freq)      | 1000                                   |

Table 4.1: Policy network configuration and PPO trainer hyperparameters used for training the stick-removal agent (from the ML-Agents YAML configuration)

**Monitoring the training process:** Training progress was monitored using the summaries generated by ML-Agents. Figure 4.3 shows the cumulative reward over training steps during PPO training. Overall, the curve trends upward, indicating that the agent increasingly learned to remove sticks while keeping the marble structure stable (i.e., receiving higher rewards due to fewer drops). It provides a compact indicator of the improvement in the policy over time.

### 4.3 Explanation Generation

Although the RL agent is trained to efficiently choose a stick that causes the fewest marbles to fall down, the reasoning behind each individual stick selection is still unknown to the user. To address this challenge, post-hoc explanation methods were used to justify these local decisions and to make the agent’s behavior more interpretable. In this work, local explanations were generated using SHAP [11], which computes a contribution value, referred to as SHAP values, and assigns them to each input feature to show how much each input feature contributes to the model decision relative to a baseline. A SHAP value can be positive or negative: positive values support a particular action, while negative values argue against it.



Figure 4.3: Cumulative reward over training steps during PPO training; higher reward indicates episodes with fewer marble drops

In this work, Unity and Python worked together as a closed loop. Unity was responsible for setting up the game environment, running simulations, and handling user interactions. Python was responsible for loading the trained agent model, performing inference, and computing the explanations using the SHAP library. Unity and Python communicated using the Unity ML-Agents Python API and side channels. This allowed explanation generation logic to be separated from Unity’s rendering loop while still enabling real-time visualization during gameplay.

During inference mode, the Python script initialized the Unity environment through the ML-Agents Python API and waited until Unity sent a signal that the scene was ready. Once the environment was ready, Unity provided the current observation vector to Python. In addition, Python also maintained an action mask that encoded which sticks were available for execution. Python then combined the observation and action mask and passed them to the trained agent model. The model outputted a selected action, representing which stick the agent would pull out next. The action mask was updated after a stick was selected and pulled out so that the removed stick was marked as unavailable.

To compute local explanations, SHAP requires a reference dataset that represents "typical" states. In our implementation, we used training observation vectors and the corresponding action masks as background dataset. Based on this background data, SHAP compares the current state against the background data to estimate a baseline model output and compute contribution values (SHAP values) that explain how the current state shifts the agent toward or away from each action.

SHAP produces explanations for each action. In our case, each stick corresponds to one action, and SHAP returns feature contributions for every stick action. When the user hovered over a particular stick, we used the SHAP values of that hovered stick’s action to explain which input features in the current state supported or opposed selecting that hovered stick.



The agent input observation vector was feature-based: each marble was represented by three position features ( $x, y, z$ ), and each stick was represented by seven features (position  $x, y, z$  and rotation  $x, y, z, w$ ). SHAP, therefore, assigned values at the feature level to both sticks and marbles. To obtain an intuitive object-level explanation, we aggregated feature-level SHAP values into one contribution score per object:

- Marble contribution = sum of SHAP values for its ( $x, y, z$ ) position features.
- Stick contribution = sum of SHAP values for its seven pose features.

We used these object-level SHAP values to generate explanation mapping for visualization. Once all mappings were generated, Python sent the explanation mappings for all sticks to Unity in a single aggregated message via a custom side channel. Unity temporarily paused agent execution while waiting for user interaction. When the user hovered over a stick, Unity looked up the hovered stick index in the received mapping and visualized the corresponding explanation (either color highlighting or shaking). This design reduced communication overhead because the system did not need to request new explanation data from Python on every hover event.

Additionally, when a stick was pulled out, sticks remained static and only marbles were affected: they might move their position and come to rest on other marbles or sticks, or they fell through the bowl opening into the tower onto the base. As a result, visually highlighting or animating marbles provides a more meaningful and intuitive representation of stability and risk in the game. On the other hand, highlighting or animating sticks would not provide meaningful cues to users. Based on the marble-level contributions, we developed two distinct visual explanation styles: a colored marbles explanation and a shaking marbles explanation.

### 4.3.1 Colored Marbles Explanation

In the colored marbles explanation, when a user hovered over a stick, all marbles were visually colored according to its corresponding SHAP values. Marbles with positive SHAP values were colored in different shades of green, representing stability. Darker green indicates a stronger positive contribution, and it supports pulling out the hovered stick. Whereas lighter green indicates a weaker positive contribution. We used the green color because it is commonly perceived as calming, lowering emotional and physiological arousal [28]. In contrast, marbles with negative SHAP values were colored in different shades of red, representing instability. Darker red corresponds to a stronger negative contribution, and it suggests a high risk of marbles falling down, whereas lighter red indicates moderate instability. The red color is often perceived as an arousing, tension-inducing color and linked with negative emotions and risk environments [28]. Overall, it showed how strongly each marble contributed to pulling out the hovered stick. Figure 4.4 shows an example of the colored marbles explanation.

### 4.3.2 Shaking Marbles Explanation

In the shaking marbles explanation, SHAP values were visualized using animation. So when the user hovered over a stick, marbles with positive SHAP values were not shaking. This means that these marbles encouraged to pull the hovered stick. In contrast, marbles with negative

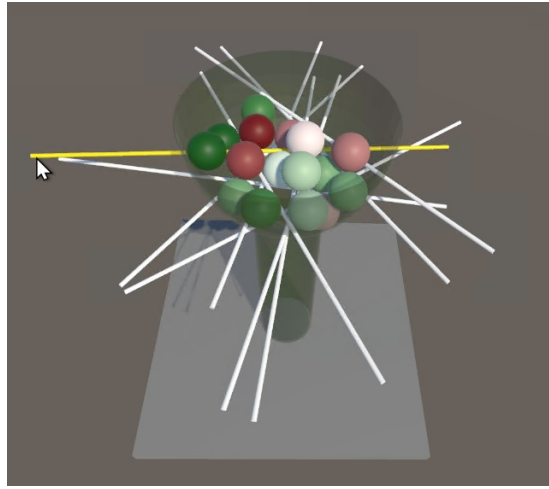


Figure 4.4: Colored marbles explanation using SHAP values

SHAP values started shaking. The intensity of the shaking was proportional to the SHAP values, which means marbles with higher negative influence exhibited stronger shaking. The core idea was to convey marble instability. When some marbles started shaking, then it meant that these marbles may fall down, shift onto other marbles or sticks, or trigger movement in nearby marbles, causing them to fall or shift to new positions.

## 4.4 User Interface

The user interface was designed to support effective interaction with the explainable RL agent. Figure 4.5 shows the complete user interface used during the user study.

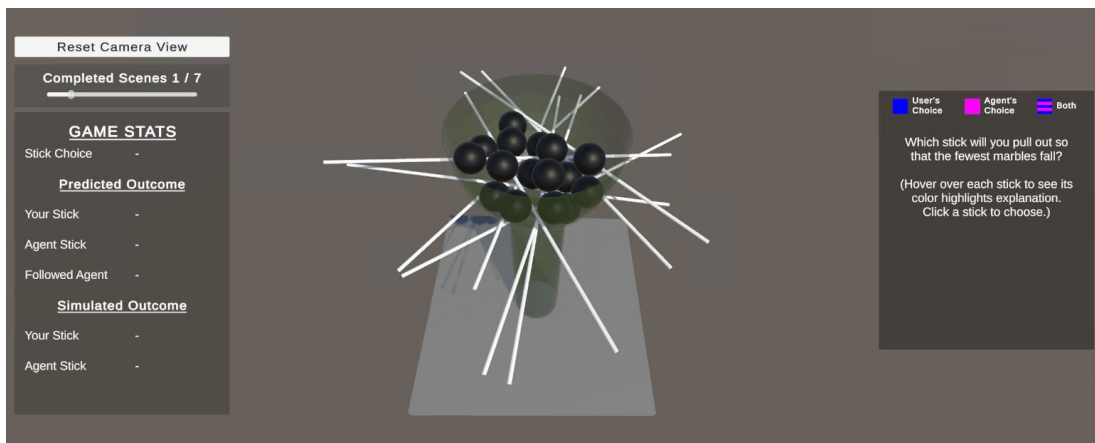


Figure 4.5: User interface of the game developed in Unity for the user study

The left side of the interface displays the game statistics, such as stick choice, follow decision, predicted outcomes, and simulated outcomes. This panel allowed participants to keep track of

their decisions and simulated outcomes and served as the reference throughout the interaction. A progress bar was shown above the game statistics panel. It tracked the number of scenes completed by the user in one explanation condition.

The right panel was used for displaying task instructions via questions, decision prompts, and a trust meter. This panel guided participants throughout the study. A small legend at the top of this panel indicated the selection status of sticks: blue marked the user’s choice, pink marked the agent’s choice, and a blue–pink striped pattern marked cases where both selected the same stick. Although not shown in the legend, yellow was used to highlight a stick when a user hovered over it, orange indicated the optimal stick (best stick), an orange–blue striped pattern indicated the user’s choice was optimal, an orange–pink striped pattern indicated the agent’s choice was optimal, and an orange–blue–pink striped pattern indicated both agent and user selected the same optimal stick.

To enable spatial understanding of the 3D game environment, users were allowed to rotate the camera freely by holding and dragging the right mouse button. This allowed participants to inspect the game from different viewpoints before making any decision. A *Reset Camera View* button is also integrated into the interface to allow participants to restore the default viewpoint at any point during the interaction.

We also aligned our explanation design and explainable user interface with the guidelines proposed by Xu et al. [6]. Explanations were always accessible to the user upon request via hovering over a stick (G1 and G2). This allowed participants to decide when they wanted to see the explanation and for which stick without interrupting the gameplay with unnecessary explanations. Furthermore, both explanation styles focused on answering *why this stick was used* to understand the agent’s decision (G4). They showed how each marble contributed to the agent’s stick choice.

In addition, explanations were aligned with the agent’s goal of choosing a stick that causes the minimum number of marbles to fall and the user’s goal of understanding the consequences and making predictions (G3). However, explanations were not tailored to specific user profiles (G3). This ensures experimental control and comparability among participants. Similarly, we partially followed G5. Users were able to explore explanations for alternative actions (sticks), but were unable to get deeper explanations.

Since the output modality of the agent is the visual action of pulling out a stick in 3D space, we provided explanations in the same modality via visual clues like coloring or shaking of marbles embedded implicitly in same space (G6 and G8). Even though authors [6] suggested using the textual explanation style as a primary format to explain the reasoning of the agent’s decision, we intentionally deviated from this guideline, as textual explanations would disrupt the immersion and interaction flow during the gameplay. Instead, we used simple, intuitive, and easily interpretable visual cues to explain the agent’s action. Table 4.2 summarizes how the interface was aligned with XAI design principles proposed by Xu et al. [6].

Table 4.2: Alignment with the XAI design guidelines proposed by Xu et al. [6]

| Guideline | Status             | Implementation in this work                                                     |
|-----------|--------------------|---------------------------------------------------------------------------------|
| G1        | Followed           | Explanations always accessible via hovering                                     |
| G2        | Followed           | User-triggered explanations via hovering                                        |
| G3        | Partially followed | Aligned with system’s and user’s goals but not personalized as per user profile |
| G4        | Followed           | Focus on concise “why” explanations                                             |
| G5        | Partially followed | Exploration across actions, not deeper explanation details                      |
| G6        | Followed           | Visual explanations match visual agent output                                   |
| G7        | Not followed       | Text avoided to preserve immersion and reduce load                              |
| G8        | Followed           | Implicit visual cues embedded in the environment                                |

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## Chapter 5

# USER STUDY AND RESULTS

This chapter begins with explaining how the task was constructed in section 5.1, the development of trust classification logic in section 5.2, and how the study was designed in section 5.3. Next, section 5.4 describes the procedure of the study. It is followed by discussing how the data that was collected by the system in section 5.5. Finally, the hypotheses, the statistical analysis, the quantitative results obtained from the user study, and the qualitative results that were obtained from the questionnaires are discussed in section 5.6.

### 5.1 Task Construction

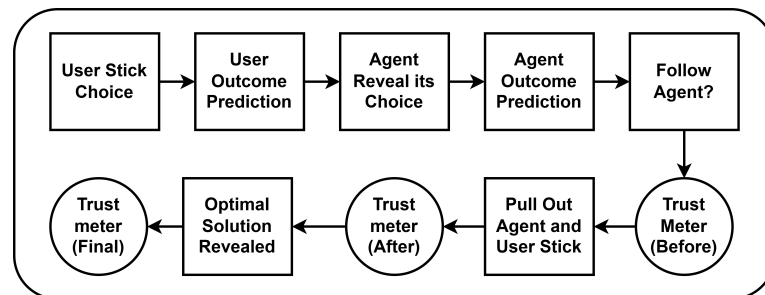


Figure 5.1: Interaction flow within a single game scene

We designed the study to evaluate how users interact with an agent who provides decision support via explanations during the gameplay. Their task was to choose a stick that they believed would result in the fewest marbles falling. Participants were told that they would work with different explainable agents which correspond to different explanation conditions. The agent would recommend actions and it might not always be optimal. They were free to either choose to follow the agent's recommendation or ignore it and follow their own decision. This encouraged the participant to treat each explanation condition independently.

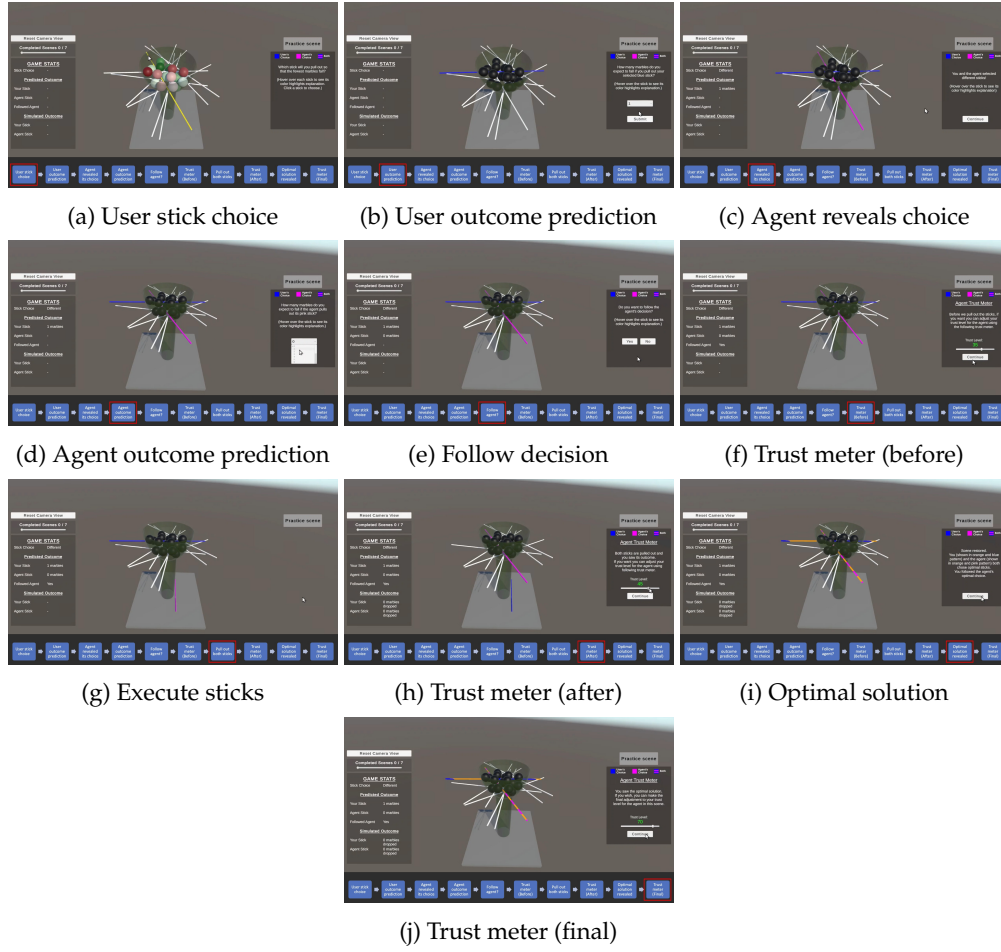


Figure 5.2: User interface screens corresponding to each step in the interaction flow shown in Fig. 5.1

In each study condition, participants played a series of scenes. Figure 5.1 shows the flow of interaction within a single game scene, and figure 5.2 shows the user interface screens corresponding to each step in the interaction flow shown in figure 5.1.

In each scene, participants observed the initial setup of the sticks and marbles. Depending on the study condition, they could hover over the sticks to see the agent’s explanation for pulling out the hovered stick (shown in Figure 5.3). The hovered stick was shown in yellow color. The explanations were available throughout the game scene. Participants could request the explanations by hovering over the stick. They were also allowed to rotate the camera to see the game from different viewpoints. This enabled participants to examine the available information before making a decision at every step.

With the help of these explanations, participants were first asked to make an initial decision of selecting a stick that they would want to pull out. Figure 5.4 shows the users’ stick choice in blue color, and they also predicted how many marbles would fall for their stick choice.

Once the participants had chosen a stick and predicted the outcome, the agent revealed its

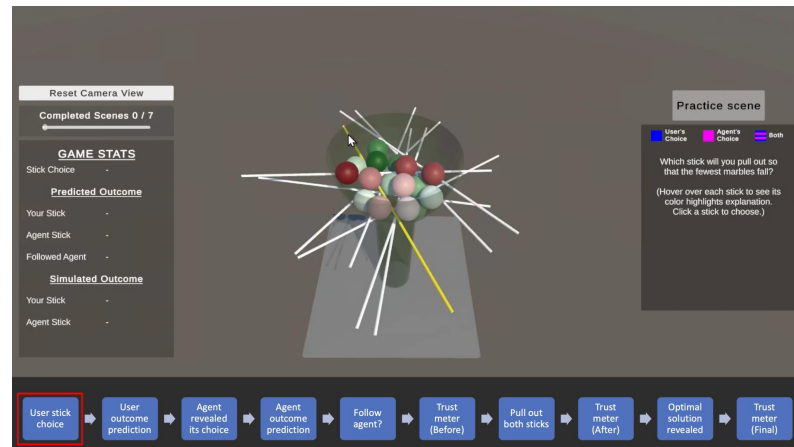


Figure 5.3: User stick choice

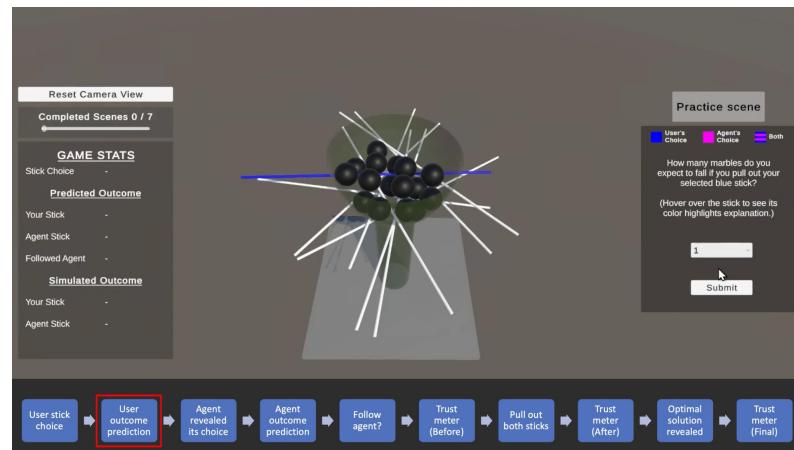


Figure 5.4: User outcome prediction

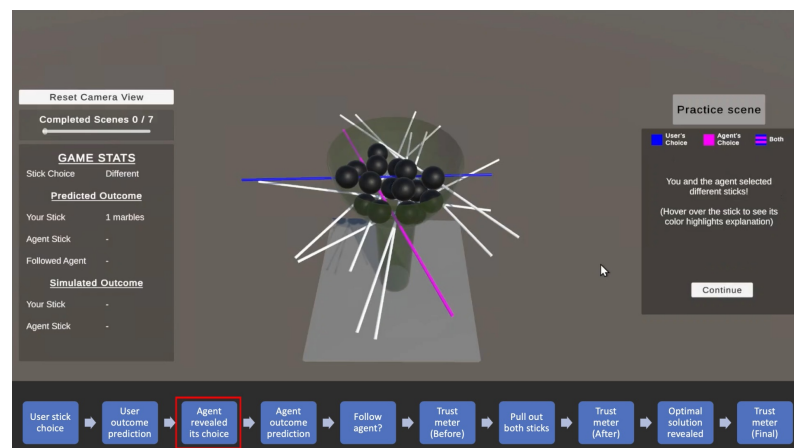


Figure 5.5: Agent reveals choice

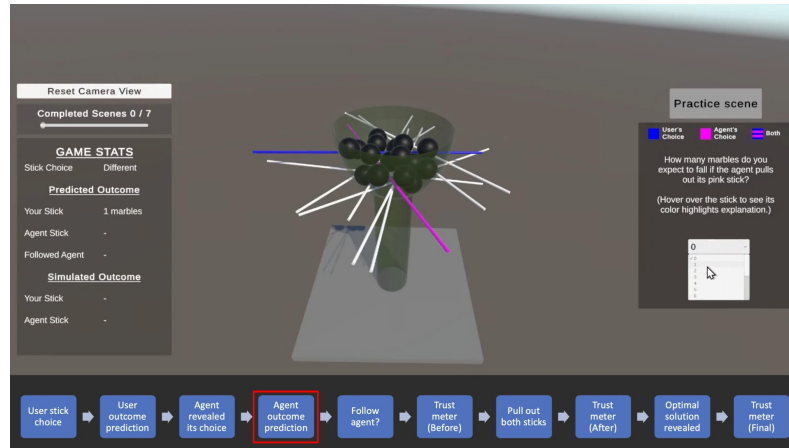


Figure 5.6: Agent outcome prediction

stick choice. Figure 5.5 shows the agent’s stick choice in pink color. Participants could inspect the explanation for the agent’s recommendation. If the participant’s choice and the agent’s recommendations differed, then participants were asked to predict the outcome of the agent’s stick choice (shown in Figure 5.6). Based on the outcome prediction for both sticks shown in the game statistics panel, participants were asked to decide whether they would follow the agent’s recommendation or ignore it and proceed with their choice (shown in Figure 5.7).

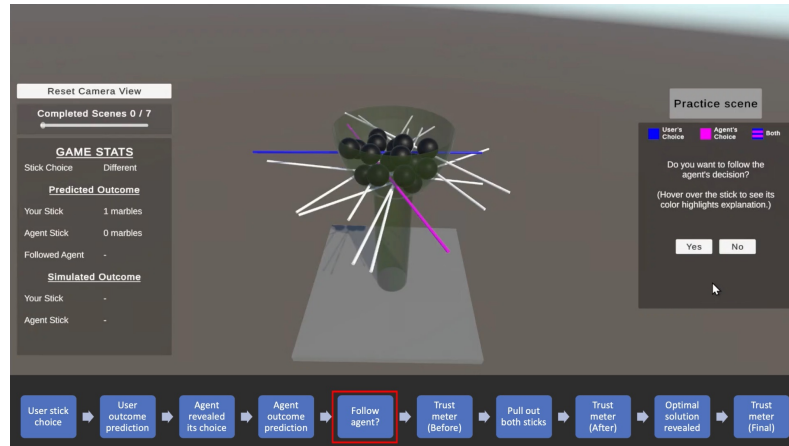


Figure 5.7: Follow decision

To build appropriate trust, they were encouraged to follow the agent when they believed that the agent was correct and proceed with their own decision when they believed the agent’s recommendation was incorrect. If both the user and the agent chose the same stick, then we assumed that participants would follow the agent’s recommendation.

Before we showed the simulation of both sticks, they adjusted their trust in the agent on the trust meter based on the recommendation and explanation provided by the agent. Figure 5.8 shows the agent trust meter slider on the right panel.

After participants reported their initial trust in the agent, we first showed the execution of



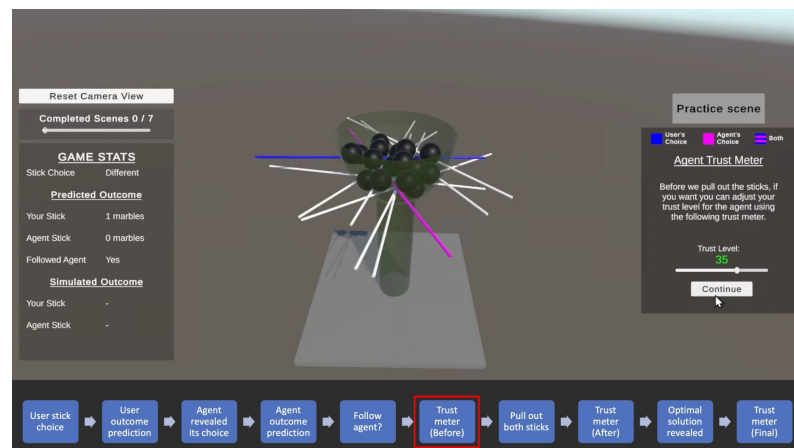


Figure 5.8: Trust meter (before)

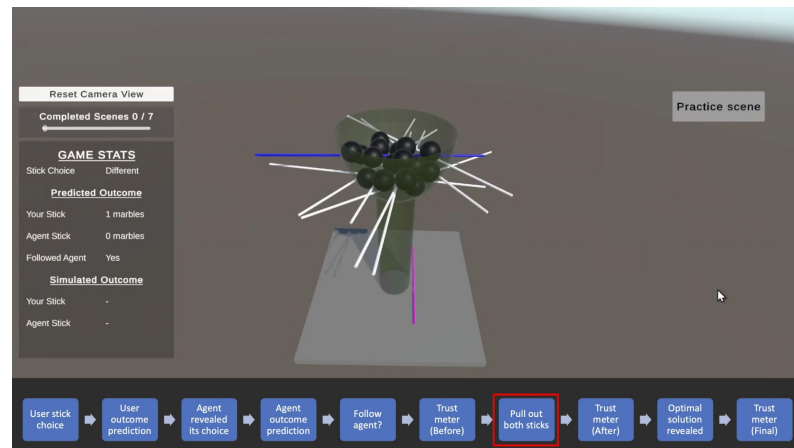


Figure 5.9: Execute sticks

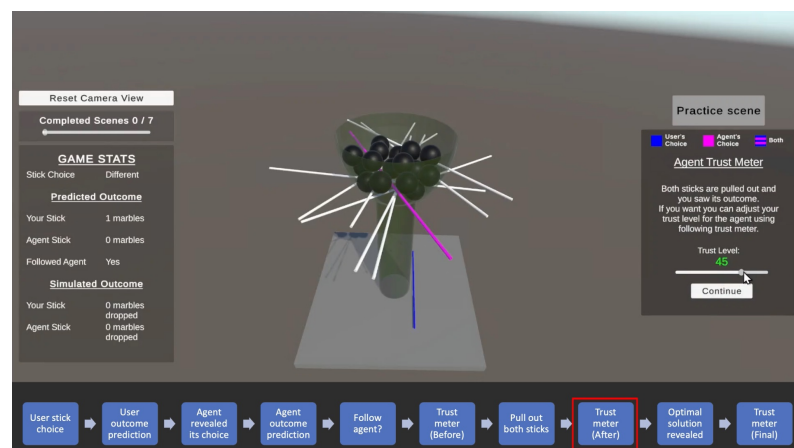


Figure 5.10: Trust meter (after)

agent's stick and the resulting movement of marbles (see Figure 5.9). Then we showed the execution of users' stick and asked participants to again report the trust in the agent using the trust meter (see Figure 5.10). This reflects the change in trust based on the explanation and the observed consequences of the actions.

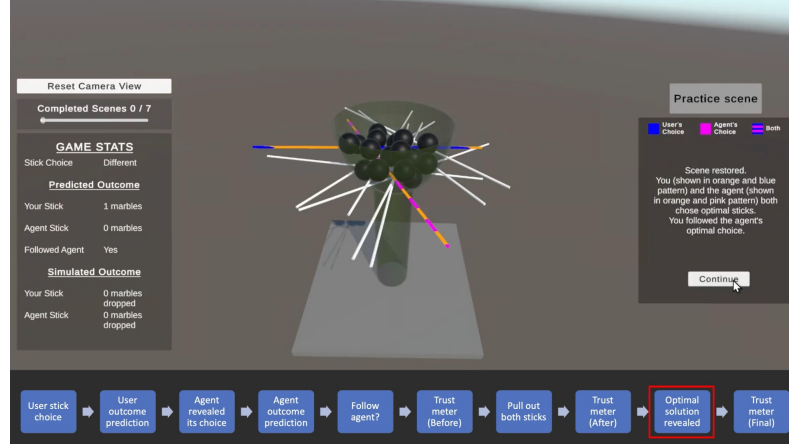


Figure 5.11: Optimal solution

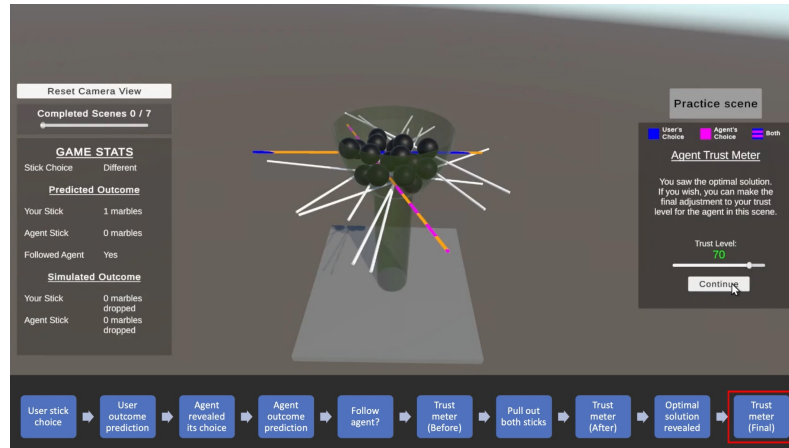


Figure 5.12: Trust meter (final)

Following the execution phase, we also showed the optimal stick that dropped the fewest marbles as the ground truth reference for each scene. In our study, the optimal stick always dropped zero marbles in all the game scenes that were selected. Figure 5.11 shows the optimal solution in orange color. Since both the agent's and the users' sticks were optimal, they were shown in orange-pink and orange-blue striped patterns, respectively. Considering both the observed outcome and the optimal solution, participants reported the final change in trust in the agent using the trust meter (see Figure 5.12).

This interaction flow was repeated for all the scenes. Showing the simulations of both sticks and the optimal solution allowed us to observe the change of trust during the scene. Lastly, following the agent's recommendation and the outcome of both sticks helped us to calibrate the trust and classify it as appropriate trust, overtrust, and undertrust.

## 5.2 Trust Classification Logic

This section deals with measuring the objective trust in the XRL agent. We developed a trust classification logic that first compared the actual outcomes of sticks. Further, we also checked the participant's follow decision: if participants rejected the agent's recommendation, then it would imply that they believed their own choice was better. Based on this concept, we labeled the participant's behavior as appropriate trust, overtrust, and undertrust for each scene using the logic shown in figure 5.13.

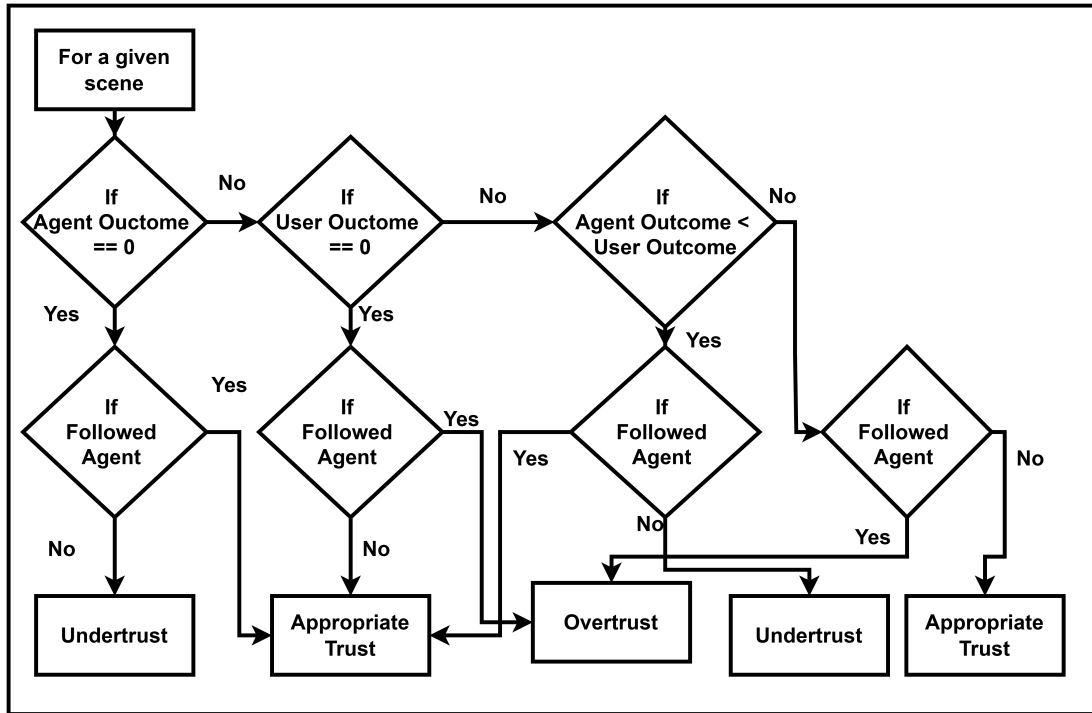


Figure 5.13: Trust classification logic adopted in this thesis

Since we measure trust in the agent, we first check the outcome of the agent's stick. If the agent selected the optimal stick, which caused zero marbles to fall, then

- Following the agent was classified as appropriate trust, as the participants correctly relied on the agent's decision.
- Not following the agent was classified as undertrust, indicating that participants unnecessarily rejected a correct optimal solution.

Next, when the agent chose a non-optimal stick, which caused marbles to fall down, and the user chose an optimal stick, then

- Not following the agent was classified as appropriate trust, as participants rejected an incorrect recommendation.

- Following the agent was classified as overtrust, as participants chose a suboptimal solution.

Lastly, when both agent and user failed to choose an optimal stick, we compared the outcome of both sticks. The stick that caused fewer marbles to fall was considered better. If the outcome of the agent's stick is better than the user's stick, then

- Following the agent was classified as appropriate trust
- Not following the agent was classified as undertrust

Similarly if the agent's outcome was equal to or worse than the user's outcome, then

- Not following the agent was classified as appropriate trust
- Following the agent was classified as overtrust

### 5.3 Study Design

We conducted a user study to evaluate whether different visual explanation styles influence user-centric aspects such as trust and satisfaction when interacting with an XRL agent in a 3D game environment. The study followed a complete within-subject study design in which each participant experienced all three explanation conditions.

We recruited participants by creating posters, placing them on information boards throughout the campus, DFKI, Saarbrücken and in student dormitories and posting them digitally on social media. A total of 18 participants (age range = 21-29, Mean (M)=24.7, Standard Deviation (SD)=1.74) took part in the study. 22% of our participants were females. We explicitly asked participants whether they had any form of color blindness or color vision deficiency and none of the participants reported having color vision impairment. This was necessary to guarantee that all participants could perceive and interpret the colored marbles explanation style as it conveys the information of stability and instability of marbles via color cues.

In this study, we investigated three study conditions: no explanation, colored marbles explanation, and shaking marbles explanation, as shown in figure 5.14. Each study condition consisted of 1 practice scene and 6 main scenes. Every scene followed the same sequence of interaction steps shown in figure 5.1.

We included a practice scene at the start of each condition. This allowed participants to get familiar with game mechanics, the user interface, user controls for rotating the camera in the game environment, the agent's behavior, and its corresponding explanation style. The data gathered from the practice scenes were not included in the statistical analysis.

The six main scenes were intentionally balanced to include both reliable and imperfect agent recommendations. There were four *good* scenes where the agent's recommended stick was the optimal stick that resulted in dropping zero marbles. There were two *bad* scenes where the agent's stick caused marbles to fall down. Thus, the agent provided an optimal recommendation

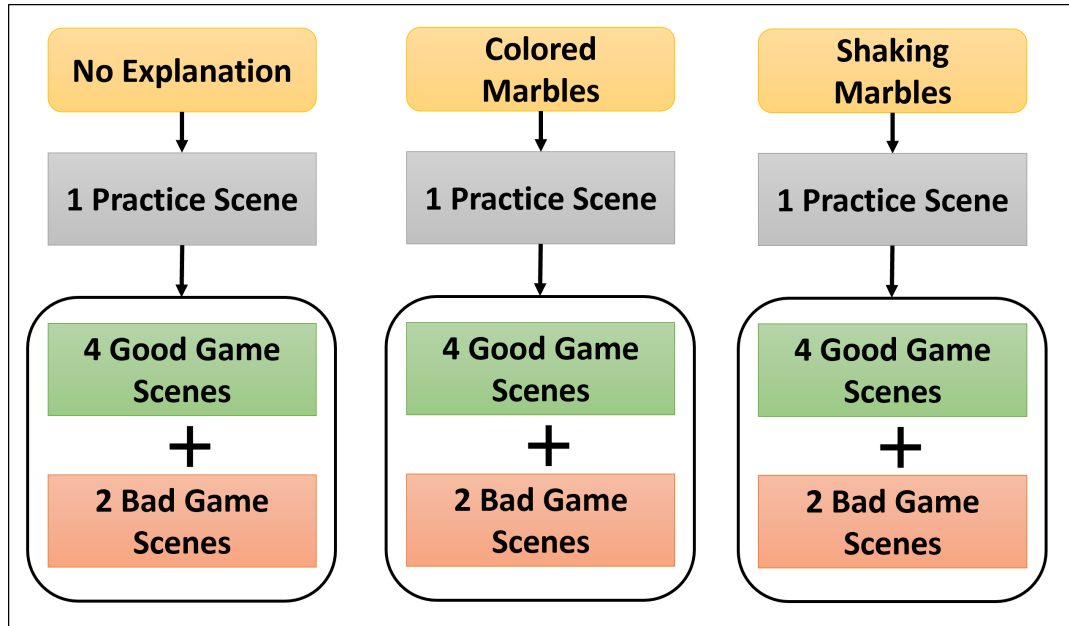


Figure 5.14: Distribution of scenes in each study condition

in 67% of the main scenes (4 out of 6). Moreover, all 3 practice scenes were intentionally designed as good scenes to boost the trust in the agent.

To avoid ordering effects and learning effects, we used a Latin square balancing design [29] to counterbalance the order of both conditions and scenes. This design ensured that each condition appeared in different orders across all the participants and each scene was presented under different explanation styles across participants.

## 5.4 Procedure

Upon arrival, participants were provided with an information sheet that described the purpose of the study and the task they would perform. After reading the sheet, they first signed the consent form and filled out the pre-study questionnaire, which covered the basic background and demographic questions. Participants then received a brief explanation of the game mechanics and the overall study flow. At the start of each study condition, they were provided information about the corresponding explanation style and how to interpret it. Participants then started the study, finishing 1 practice scene and 6 game scenes under each study condition. At the end of each study condition, participants filled out post-condition questionnaires assessing their trust and satisfaction with the explanation style. All participants took a minute break in between study conditions. After completing all 3 study conditions, participants completed a post-study questionnaire. Detailed information about the questionnaires can be found in the appendix.

## 5.5 Data Collection

During the user study, we collected both objective data and subjective self-reported measures. The following objective data were logged by the user interface during the game scene.

- We recorded the participant's stick choice and the participant's predicted outcome for that stick.
- We also recorded the agent's recommended stick and the participant's predicted outcome for the agent's stick.
- We logged whether the participant chose to follow the agent's recommendation or their own decision.
- Lastly, we logged the simulated outcome for both the agent's and the participant's stick choice (i.e., the number of marbles that dropped after pulling out both sticks).

Subjective data were collected in two ways:

- Participants adjusted the trust meter and reported their trust in the agent at 3 distinct phases within each game scene: Before the execution phase, after the execution phase, and after the optimal solution was revealed. All trust meters were logged by the user interface. These values allowed us to capture the changes in agent trust across the interaction flow.
- At the end of each study condition, participants completed questionnaires assessing their subjective experience with the corresponding explanation style. To measure the user satisfaction with the explanations, we used the Explanation Satisfaction Scale proposed by Hoffmann et al. [30]. We selected this scale because it is specifically designed to evaluate the satisfaction with the explanations in XAI setting. It captures key attributes such as understandability, feeling of satisfaction, sufficiency of detail, completeness, usefulness, accuracy, and trustworthiness [30].
- After completing all the study conditions, participants filled out a final post-study questionnaire that included open-ended questions. They ranked all explanation styles and provided qualitative feedback by comparing them. We used these open-ended responses to complement the quantitative findings to gain deeper insights from the study.

## 5.6 Analysis and Results

This section presents the statistical analysis and quantitative results of the user study. We framed our research questions focusing on the user trust calibration, subjective dynamics of changes in trust across 3 distinct phases, and user satisfaction under different visual explanation conditions. The resulting research questions are as follows:

- **RQ1:** How do different visual explanation styles influence the users' trust in the agent's recommendations?

- **RQ2:** How do different explanation conditions influence the direction of the users' trust across the three phases within a game scene (before execution, after execution, and after showing the optimal solution)?
- **RQ3:** How does the agent's error severity (number of marbles dropped) affect the direction of the users' trust after executing the agent's recommended stick?
- **RQ4:** How do different visual explanation styles impact the user satisfaction during gameplay?

### 5.6.1 Hypotheses

We formulated hypotheses based on the above research questions to examine the effects of visual explanation styles on trust and satisfaction. Overall, we hypothesized that visual explanations would lead to an increase in trust and satisfaction compared to giving no explanation at all. More specifically, we stated the following hypotheses for all three RQs:

#### **RQ1: Trust Calibration**

- H1: Visual explanation styles influence users' trust calibration towards the agent's recommendation, specifically:
  - H1a: Visual explanations increase appropriate trust compared to the no-explanation condition.
  - H1b: Visual explanations reduce overtrust compared to no-explanation condition.
  - H1c: Visual explanations reduce undertrust compared to no-explanation condition.

#### **RQ2: Direction of Trust Change Across Phases**

- H2: The explanation condition has an effect on how users update their subjective trust across three phases, specifically:
  - H2a: Explanation conditions lead to different subjective trust updates after being shown the execution of both sticks.
  - H2b: Explanation condition leads to different subjective trust updates the optimal solution is shown.
  - H2c: Explanation condition leads to different overall trust updates across the entire scene.

#### **RQ3: Trust Change in Correlation to Outcome of Agent's Stick**

- H3: The trust change after the execution of the agent's recommended stick differs across error severities (0, 1, 2 marbles dropped); Specifically, a higher error severity leads to a stronger decrease in trust.

**RQ4: Explanation Satisfaction**

- H4: User satisfaction is not equal across the three explanation styles (none, color, shake); at least one condition differs from another.

**5.6.2 Statistical Approach**

Since all participants experienced all three explanation conditions: no explanations (none), colored marbles explanation (color), and shaking marbles explanation (shake), we conducted all analyses using a within-subject design. For each dependent variable, we first tested for an overall effect of the independent variable using repeated-measures ANOVA [31]. Then we assessed the normality of ANOVA residuals using the Shapiro-Wilk test [32]. When the assumption of normality was violated, we additionally performed the nonparametric Friedman test [33]. Further, we conducted post-hoc comparisons between different pairs of independent variables using paired Wilcoxon signed-rank tests [34]. Since the study involved three conditions and a small number of pairwise comparisons, we used unadjusted p-values in the main results and figures [35, 36].

We included all 18 participants in the analysis because there was no participant who did not understand the task and failed to follow the study. The detailed quantitative results are as follows.

**5.6.3 Quantitative Results**

**RQ1: How do different visual explanation styles influence the users' trust in the agent's recommendations?**

**Method:** We used three trust calibration measures to quantify user trust: appropriate trust, overtrust, and undertrust. We annotated each trial with a trust label using the trust classification logic (see Section 5.2). For each participant and explanation condition, we computed the proportion of trials for each trust label.

**Results:** Figure 5.15 presents the trust calibration results across explanation conditions. The explanation conditions are shown on the x-axis, and the proportion of trials for each trust label is shown on the y-axis. Each point represents one participant's proportion of trials in that trust label. The figure is faceted into three panels, one for each trust label (Appropriate trust, Overtrust, Undertrust). Overall, the results indicate that explanation style significantly affected users' trust calibration.

The repeated-measures ANOVA showed a significant effect of condition on three trust labels (For *appropriate trust* ( $F(2, 33.97) = 5.14, p = .011$ ), for *overtrust* ( $F(1.75, 29.68) = 3.54, p = .048$ ), and for *undertrust* ( $F(1.99, 33.78) = 16.74, p < .001$ )). The Shapiro-Wilk test revealed that the assumption of residual normality was violated for *appropriate trust* ( $W = 0.948, p = .021$ ) and for *overtrust* ( $W = 0.941, p = .010$ ). The residuals were normally distributed for *undertrust* ( $W = 0.968, p = .165$ ). Thus, we conducted Friedman test for all three trust labels, which also confirmed a significant effect of condition on all three trust labels (for *appropriate trust*



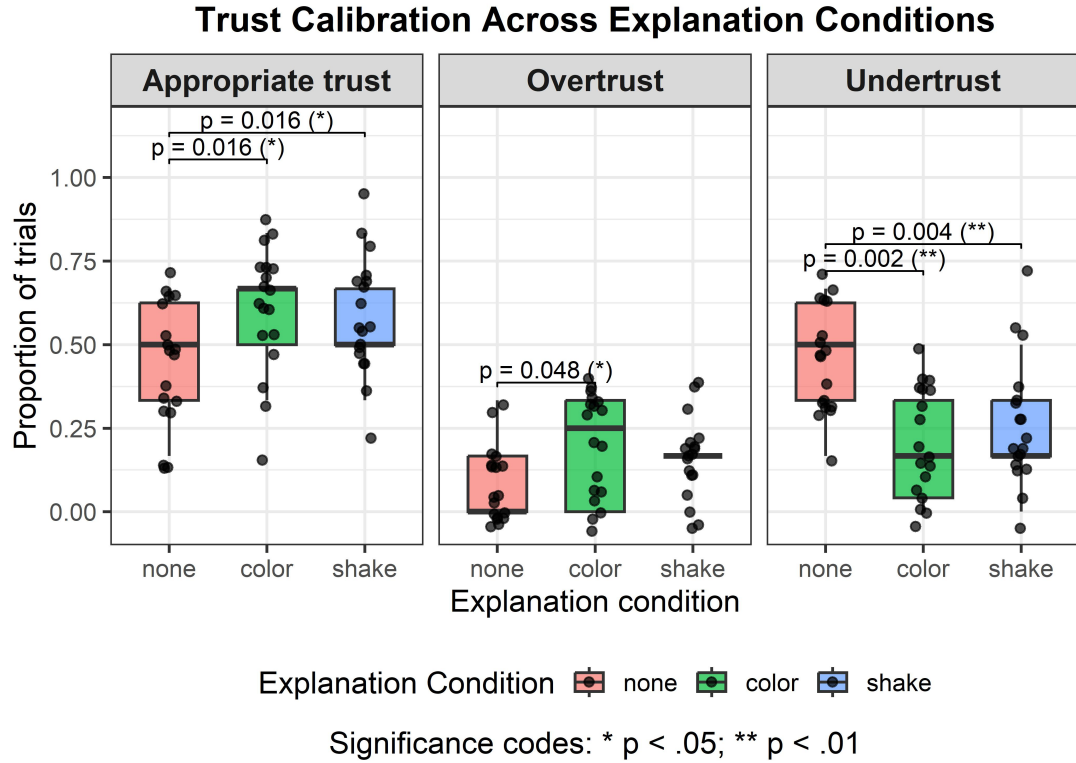


Figure 5.15: Trust calibration across explanation conditions (proportion of trials labeled as appropriate trust, overtrust, and undertrust). Brackets show paired Wilcoxon signed-rank tests

( $\chi^2(2) = 6.70, p = .035$ ) for *overtrust* ( $\chi^2(2) = 6.49, p = .039$ ), and for *undertrust* ( $\chi^2(2) = 18.29, p < .001$ ). Post-hoc Wilcoxon signed-rank tests indicated that both color and shake explanation conditions showed a significant increase in *appropriate trust* and reduced *undertrust* (see figure 5.15) over the none condition, and there was no significant difference between color and shake conditions. For *overtrust*, the color explanation significantly increased the *overtrust* over the no-explanation condition ( $p = .048$ ), while comparisons involving the shake condition showed no significant difference.

**RQ2: How do different explanation conditions influence the direction of the users' trust across the three phases within a game scene (before execution, after execution, and after showing the optimal solution)?**

Participants reported trust in the agent at three distinct points in each game scene: before the execution phase (denoted as "before"), after the execution phase (denoted as "after"), and after the optimal solution phase (denoted as "final"). The bar graph (see Figure 5.16) provides a descriptive overview of mean trust levels across three phases (before execution, after execution, and final) for each explanation condition (none, color, shake). The explanation condition is on the x-axis, mean trust levels are on the y-axis, and there are separate panels for each phase. Overall, trust tends to increase from the before-execution phase to after execution and then to the final phase under all three conditions.

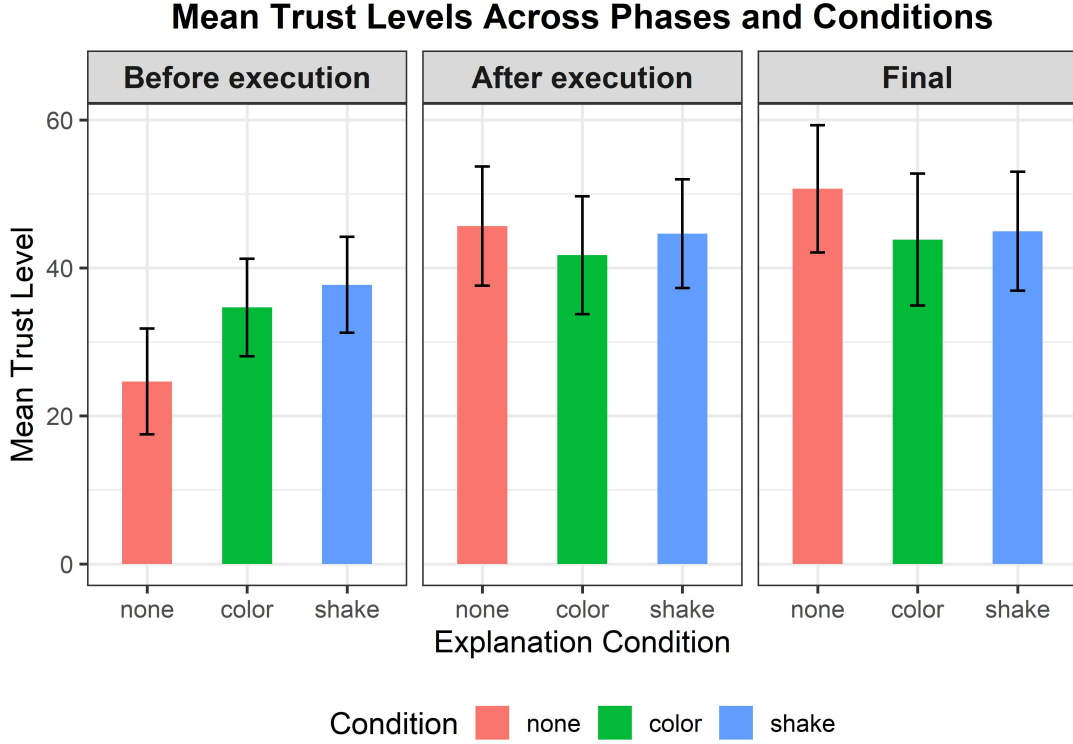


Figure 5.16: Mean trust levels across three phases (before execution, after execution, and final) and three explanation conditions (none, color, shake)

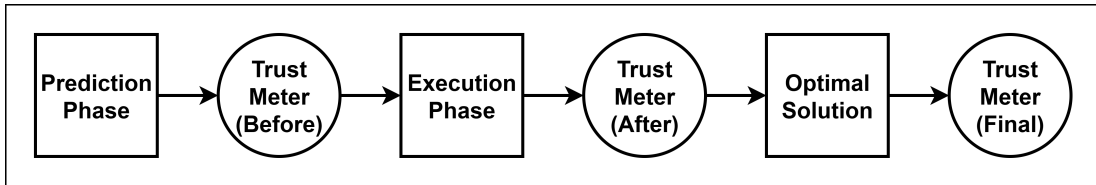


Figure 5.17: Overview of the phase structure and trust measurement points within a single game scene

**Method:** Figure 5.17 provides a high-level summary of the detailed interaction flow presented earlier in Figure 5.1. It highlights the three phases of each game scene and the three time points at which participants reported their trust (before execution, after execution, and after the optimal solution was revealed). This formed the basis for measuring the trust-change between phases. Similar to prior work on trust direction [24], we computed three deltas of trust change:

1.  $\Delta T_{\text{after-before}}$ , defined as the change of the trust in the agent after being shown the execution of the agent's and the user's stick.
2.  $\Delta T_{\text{final-after}}$ , defined as the change of the trust in the agent between the execution phase and after being shown the optimal solution.
3.  $\Delta T_{\text{final-before}}$ , defined as the overall change of the trust in the agent within the game scene.

We used signed difference instead of absolute difference to measure the direction of trust. Positive values indicate an increase in trust, whereas negative values indicate a decrease in trust. We computed the mean trust change for each participant and condition and performed the analysis for within-subject design.

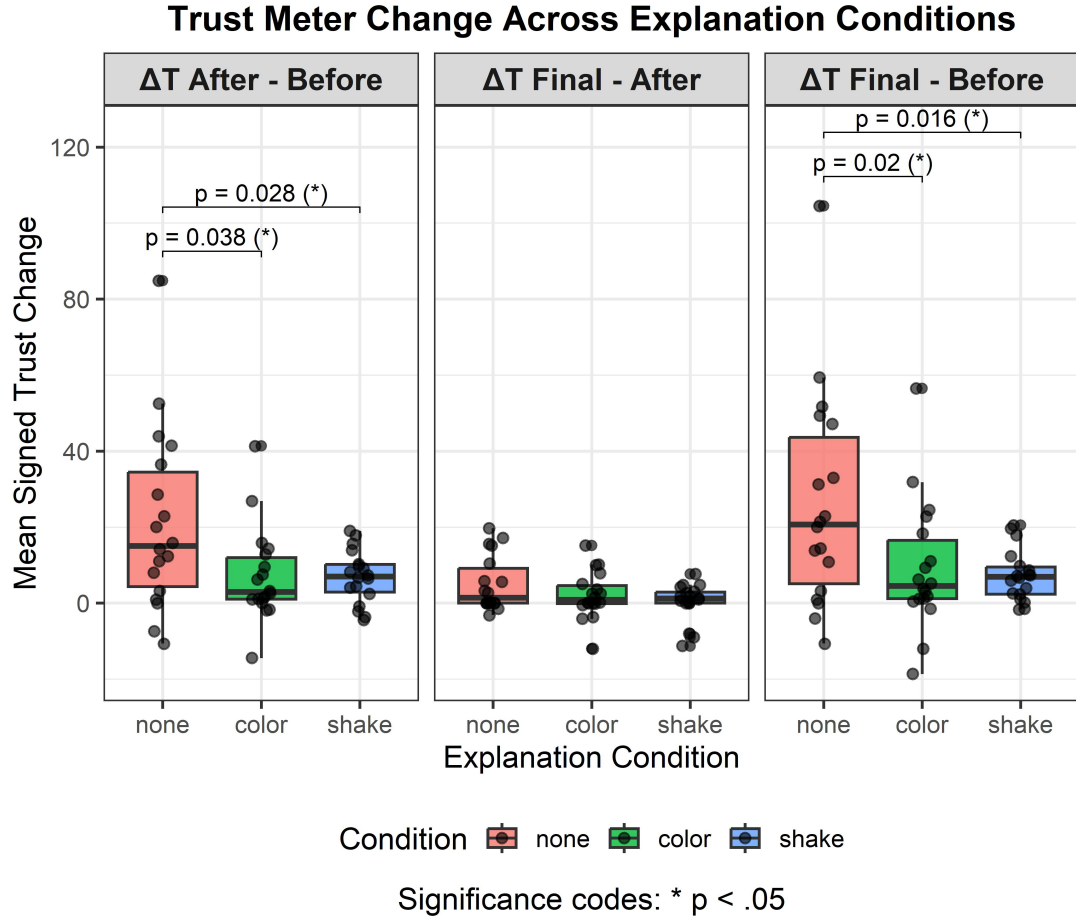


Figure 5.18: Signed trust change across explanation conditions for three phase transitions: After-Before, Final-After, and Final-Before

**Results:** We reported results in figure 5.18. The x-axis represents the explanation condition, and the y-axis represents the mean signed trust change between two phases. There are separate panels for each phase transition. Each dot corresponds to one participant's mean trust change under that condition.

For  $\Delta T_{\text{after-before}}$  repeated-measures ANOVA suggested a significant effect of condition in the trust change from before the execution phase to after the execution phase. However, the normality assumption is violated (Shapiro-Wilk:  $W = 0.906$ ,  $p < .001$ ). Therefore, we performed the Friedman test, and it confirmed a significant effect of condition ( $\chi^2(2) = 7.11$ ,  $p = .029$ ). Post-hoc Wilcoxon test revealed that trust increased more in the none condition as compared to color ( $p = .038$ ) and shake ( $p = .028$ ). There was no significant difference between color and shake condition.

For  $\Delta T_{\text{final-after}}$ , there was no significant effect of condition in the repeated-measures ANOVA ( $F(1.53, 26.06) = 3.33, p = .063$ ). The Shapiro-Wilk test indicated no violation of the normality assumption ( $W = 0.981, p = .538$ ). The Friedman test also confirmed that condition has no effect on the change in trust from after the execution phase to the final phase after seeing the optimal solution. No pairwise Wilcoxon comparison revealed any significant results.

For  $\Delta T_{\text{final-before}}$ , repeated-measures ANOVA indicated a significant effect of condition ( $F(1.46, 24.80) = 5.48, p = .017$ ). But the residual violated the normality assumption (Shapiro-Wilk:  $W = 0.894, p < .001$ ). Although the Friedman test did not show a significant effect of condition ( $\chi^2(2) = 3.11, p = .211$ ), the Wilcoxon test indicated an increase in overall trust in none compared to color ( $p = .020$ ) and shake ( $p = .016$ ). Since the omnibus Friedman test was not significant, it is suggested that this effect should be interpreted cautiously. Lastly, there was no significant difference between color and shake condition.

**RQ3: How does the agent's error severity (number of marbles dropped) affect the direction of the users' trust after executing the agent's recommended stick?**

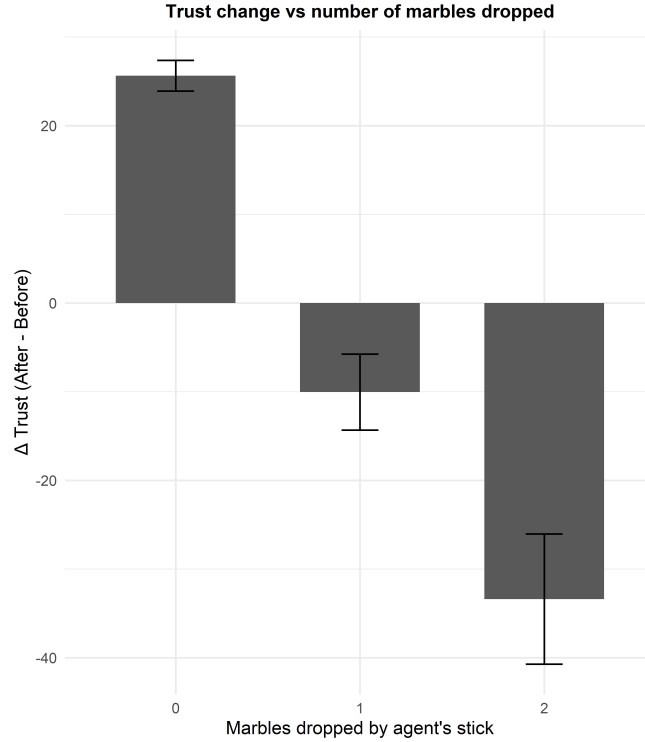


Figure 5.19: Mean signed trust change from before execution to after execution  $\Delta T_{\text{after-before}}$  of the agent's recommended stick as a function of agent error severity (0, 1, or 2 marbles dropped)

Figure 5.19 shows the initial descriptive analysis of the mean trust change from before execution to after execution ( $\Delta T_{\text{after-before}}$ ) of the agent's recommended stick across three levels of agent error severity. The x-axis represents the number of marbles dropped by the agent's recommended stick (0, 1, or 2 marbles dropped). The y-axis shows the signed trust change  $\Delta T_{\text{after-before}}$ . Each bar corresponds to the average  $\Delta T_{\text{after-before}}$  at that drop value. The figure suggests a clear

pattern: when the agent's stick drops 0 marbles, then the trust change is positive. But once the agent's stick drops 1 marble, the trust change becomes negative and decreases further for 2 marbles dropped. This indicates that a higher error severity is associated with a stronger reduction in trust change from before to after execution of the stick.

**Method:** We again used the trust meter rating that was collected before and after execution of sticks (denoted as  $\Delta T_{\text{after-before}}$ ). To analyze how the agent's error severity affected the  $\Delta T_{\text{after-before}}$ , we used the actual simulated outcome of the agent's recommended stick. We categorized each trial into three severity levels based on the number of marbles dropped by the agent (0, 1, or 2). For the within-subject study design, we aggregated the mean  $\Delta T_{\text{after-before}}$  separately for each participant and for each error severity level. This resulted in one value per participant per severity level, which was used for the repeated measures tests.

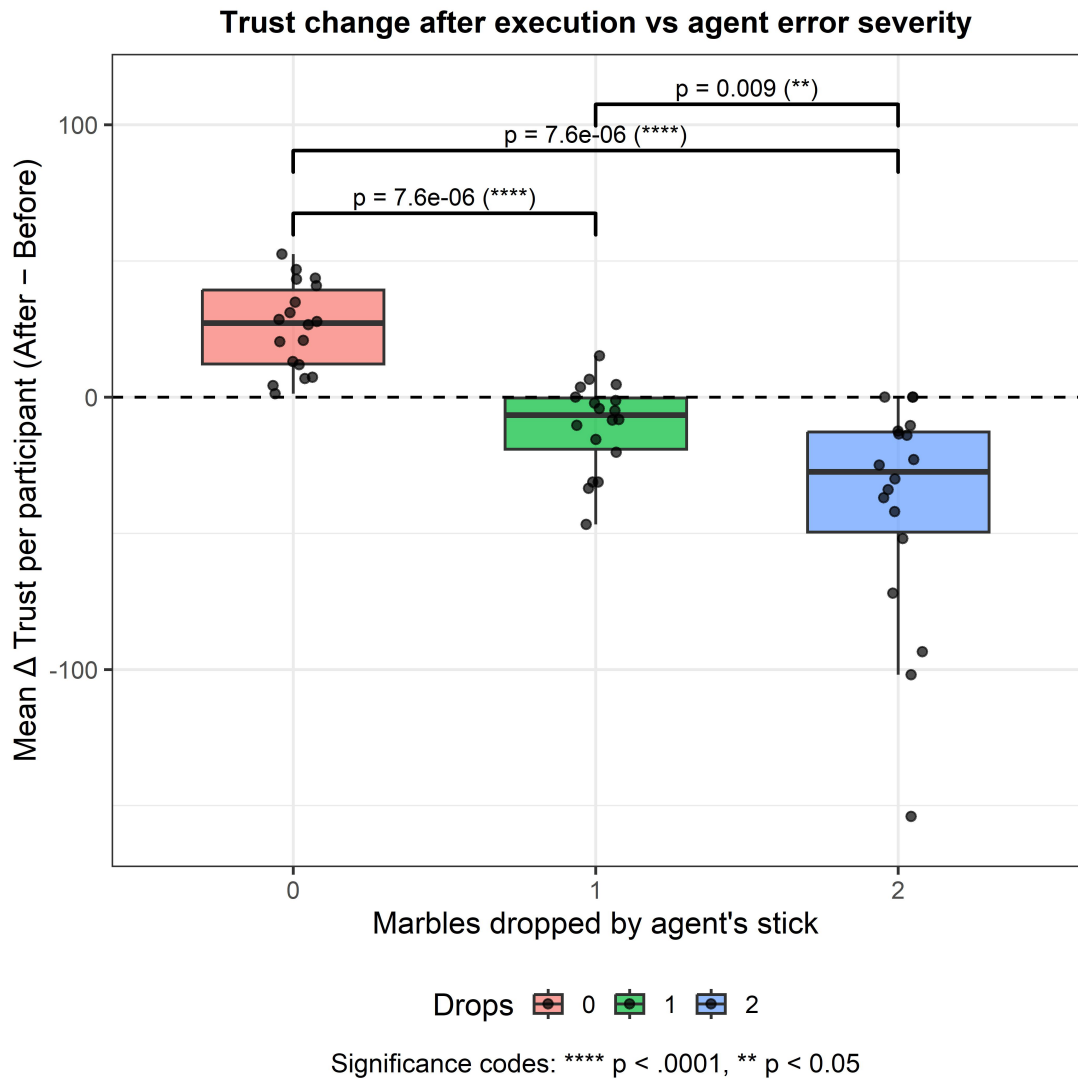


Figure 5.20: Mean signed trust change  $\Delta T_{\text{after-before}}$  across three agent outcome severities (0, 1, or 2 marbles dropped). Brackets indicate paired Wilcoxon post-hoc comparison

**Results:** Figure 5.20 summarizes the results of how trust changed with the agent’s severity. The x-axis shows the number of marbles dropped by the agent’s recommended stick, and the y-axis shows the mean signed trust change per participant. Each dot represents the one participant’s mean  $\Delta T_{\text{after-before}}$  for that severity level. A box plot visualizes the distribution across participants.

A repeated-measures ANOVA revealed a significant main effect of error severity on mean  $\Delta T_{\text{after-before}}$  ( $F(1.40, 23.80) = 21.57, p < .001$ ). However, the ANOVA residuals violated the normality assumption (Shapiro-Wilk:  $W = 0.881, p = 6.82 \times 10^{-5}$ ). Thus, we also confirmed this effect using the Friedman test ( $\chi^2(2) = 28.761, p = 5.685 \times 10^{-7}$ ) which also indicated a significant effect. Post-hoc paired Wilcoxon tests suggested that there are significant differences between all severity levels: 0 vs. 1 ( $p = 7.63 \times 10^{-6}$ ), 0 vs. 2 ( $p = 7.63 \times 10^{-6}$ ) and 1 vs. 2 ( $p = .009$ ).

To account for the repeated measurements per participant and individual baseline differences in trust change, we additionally fitted a linear mixed-effects model. We treated the trust change as the dependent variable and the number of marbles dropped by the agent’s stick as a categorical independent variable. We used 0 marbles drop as a reference and added a random-intercept per participant.

$$\Delta T_{ij} = \beta_0 + \beta_1 \mathbb{I}(\text{Drops} = 1) + \beta_2 \mathbb{I}(\text{Drops} = 2) + u_j + \varepsilon_{ij}, \quad (5.1)$$

where  $\Delta T_{ij}$  denotes the trust change for observation  $i$  from the participant  $j$ ,  $u_j \sim \mathcal{N}(0, \sigma_u^2)$  captures the participant-specific baseline differences (random-intercept), and  $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$  is the residual error term.

The intercept  $\beta_0$  represents the expected trust change when the agent’s stick drops 0 marbles. We obtained,  $\beta_0 = 25.639$ , which indicates that the average increase in trust after execution of the agent’s optimal stick. Compared to this baseline, trust change decreased substantially when the agent dropped marbles:  $\beta_1 = -35.801$  for 1 drop and  $\beta_2 = -58.681$  for 2 drops. Thus, the predicted mean (when  $u_j = 0$ ) for Drops = 1 was  $\beta_0 + \beta_1 = -10.162$  and for Drops = 2 was  $\beta_0 + \beta_2 = -33.042$ . Thus, relative to Drops = 0 ( $\hat{\mu} \approx +25.64$ ), trust change decreased for Drops = 1 ( $\hat{\mu} \approx -10.16, \beta_1 = -35.80, p < 2 \times 10^{-16}$ ) and further decreased for Drops = 2 ( $\hat{\mu} \approx -33.04, \beta_2 = -58.68, p < 2 \times 10^{-16}$ ). Model-based pairwise contrasts between drop severity levels were obtained using estimated marginal means (implemented via the R package `emmeans`). Holm-adjusted `emmeans` contrasts confirmed significant differences between 0 vs. 1 ( $\Delta = 35.8, p < .0001$ ), 0 vs. 2 ( $\Delta = 58.7, p < .0001$ ), and 1 vs. 2 ( $\Delta = 22.9, p = .0005$ ), supporting a severity-dependent reduction in trust.

#### RQ4: How do different visual explanation styles impact the user satisfaction during game-play?

**Method:** The Explanation Satisfaction Scale (ESS) showed high internal consistency (Cronbach’s  $\alpha = 0.95$ ). This indicated that all ESS items contributed positively to the overall scale, and no removal of an item improved reliability. We computed a mean satisfaction score across all ESS items for each participant and condition.

**Results:** Figure 5.21 summarizes the user’s satisfaction with explanations across all explanation

conditions. The repeated-measures ANOVA showed a significant effect of condition on mean satisfaction score ( $F(1.36, 23.08) = 25.66, p < .001$ ). The Shapiro-Wilk suggested no violation of the normality assumption ( $W = 0.974, p = 0.279$ ). We also confirmed this result with the Friedman test ( $\chi^2(2) = 17.043, p = 0.000199$ ). Post-hoc Wilcoxon tests indicated that color ( $p = 0.000374$ ) and shake ( $p = 0.000652$ ) explanations led to higher satisfaction compared to none. However, there was no significant difference between color and shake.

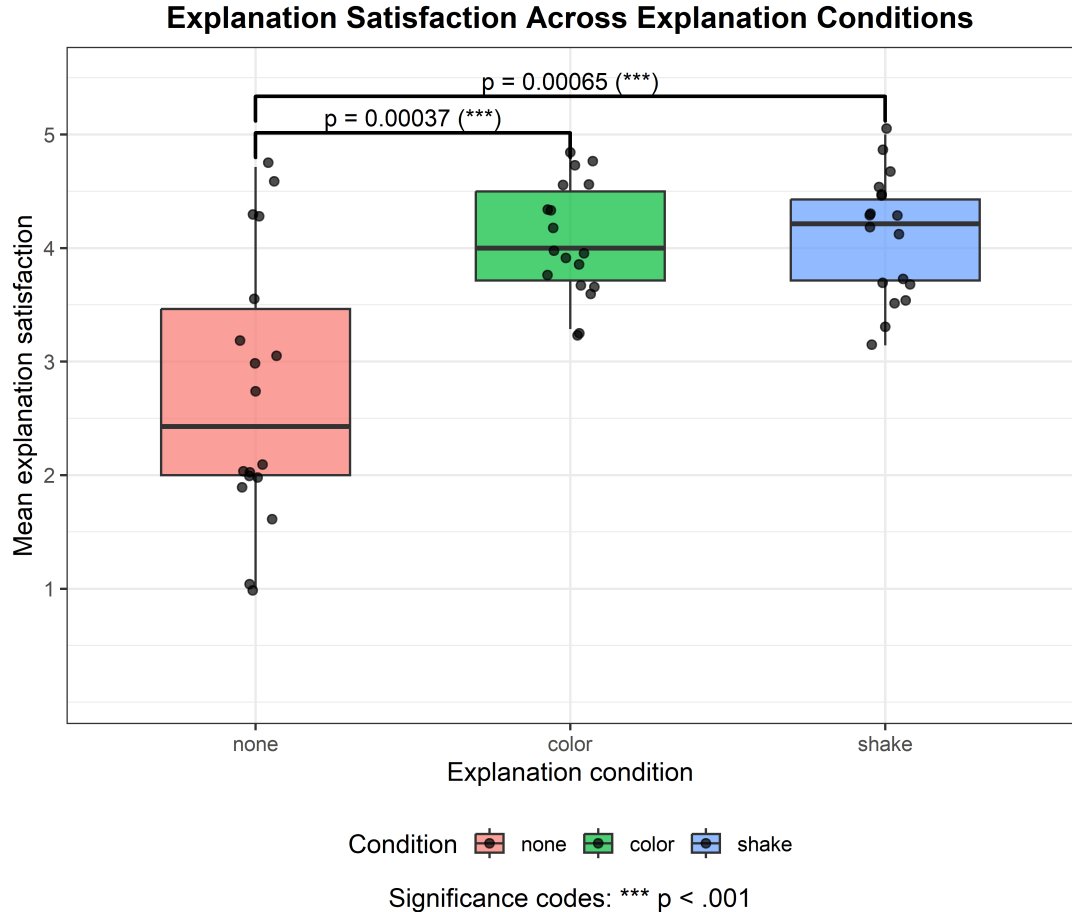
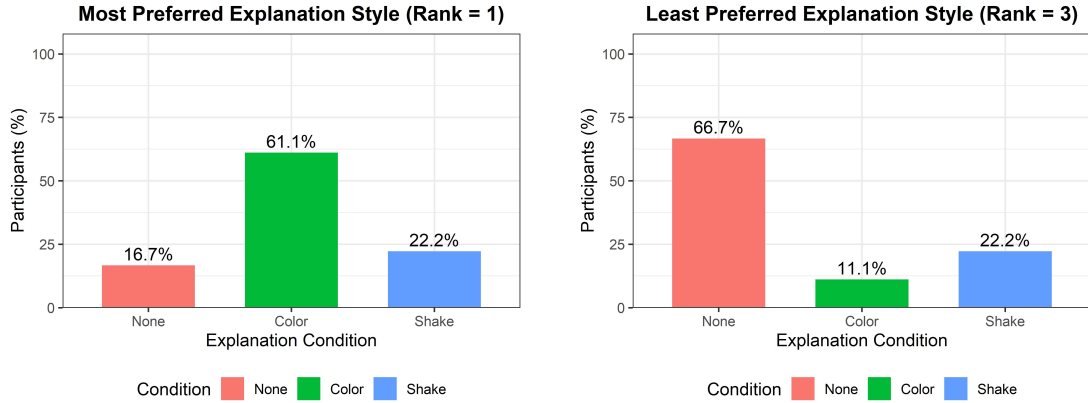


Figure 5.21: User satisfaction with the explanation style across three conditions (none, color, shake)

#### 5.6.4 Qualitative Results

To complement the quantitative analysis, we also analyzed participants's feedback for open-ended questionnaires. We collected feedback at two points: (1) after completing each explanation condition (post-condition questionnaire) and (2) after completing all three explanation conditions (post-study questionnaire).

In the post-study questionnaire, we asked participants to rank each explanation condition based on their overall preference. Rank = 1 indicates the explanation condition that the participants



(a) Most preferred explanation style (percentage of participants ranking each condition 1st)

(b) Least preferred explanation style (percentage of participants ranking each condition last)

Figure 5.22: Participant preferences across explanation conditions. Bars show the percentage of participants who ranked each condition as most preferred (a) or least preferred (b) in post-study questionnaire (N=18)

preferred the most, whereas Rank = 3 indicates the least preferred explanation condition. Figure 5.22 summarizes these rankings and reveals a clear pattern across all three conditions. In the "Most Preferred" plot (Figure 5.22a), color was ranked first by the majority of the participants (61.1%, 11/18), shake was ranked first by 22.2% (4/18) of the participants, and none by 16.7% (3/18) of the participants. The "Least Preferred" plot (see figure 5.22b) shows the opposite trend. None was most frequently ranked last (66.7%, 12/18), followed by shake (22.2%, 4/18) and color (11.1%, 2/18). Overall, these rankings indicate that the majority of participants preferred the colored marbles explanation style the most and found it enjoyable, whereas the absence of explanation was perceived as the least helpful. The shaking marbles explanation condition suggested mixed experiences and showed greater variability in perceived usefulness and clarity compared to the colored marbles explanation condition.

Along with the ranking question, we also asked participants (1) to provide feedback for each explanation condition after completing the corresponding condition and (2) to compare all the explanation styles at the end of the study. Based on the collected responses from both questionnaires, the majority of the participants described the colored marbles explanation condition as clear, intuitive, and fast to interpret. They emphasized that it was easier to assess safe and risky marbles with the green and red cues. These cues helped them to focus on the most critical marbles, which reduced the effort and supported their decision-making. For example, P02 wrote that "colors clearly show which marbles are at risk of dropping". Further, P07 explicitly liked the color explanation because he was able to see the confidence of the AI agent on each ball, and cues matched the simulated outcome when the stick was pulled. Several participants noted that the colored marbles explanation helped them to focus more on unstable marbles. Thus, color was not only visually appealing but also perceived as informative for decision-making.

Out of all three explanation conditions, None condition was the least preferred explanation style. The most consistent criticism was the reduced transparency and slower decision-making process. Participants repeatedly stated that there was no hint about why the agent chose a stick, and therefore it was harder for them to judge whether the recommendation given by the



agent was reliable or not. Thus, lack of explanation made it a little more difficult to trust the agent. For instance, P09 stated that "it was the most difficult to trust the agent since there was no reasoning backing up the decision." Overall, none was perceived as the least supportive because of the lack of explanation for the agent's reasoning.

Although there were a few participants who ranked None as the most preferred explanation style, their comments point to two main reasons. First, they preferred relying on their own judgment and observational skills rather than being influenced by an explanation. For instance, P16 stated, "Because I trusted my intuition more" and P03 also emphasized similarly that he believed in his observational skills first. Second, these participants first completed the none condition and only later experienced the color or shake condition. So they first learned to play the game based on their own intuition and observation skills without any explanation. Once they got familiar with the game and developed their own strategy and mental model of the agent, they experienced the visual explanations as misleading or distracting. In those cases, they avoided explanations and instead based their decisions directly on the physical configuration of the sticks and marbles they observed.

The shaking marbles explanation condition showed some mixed-reactions. The same amount of participants ranked it as the most preferred and the least preferred explanation style. On the positive side, some participants found shaking to be helpful and engaging for visualizing the potential movements of marbles. For example, P01 noted that "the shaking speed showed which marbles are expected to be affected the most. This was helpful for decision-making.", while P09 reported that "the vibration really helps in identifying any missed marbles in the assumption of falling." P02 also said that "it was helpful to trust as I saw the real animation". On the negative side, several participants reported that the vibrations were distracting and overwhelming. Participants found that shaking marbles complicated their judgment of the stick outcome and increased the time needed for decision-making. In short, shaking was appreciated by some participants when it acted as a clear attention cue. However, some participants disliked it when they could not reliably interpret what the motion meant.

Overall, color was preferred because it provided a stable, readable mapping of colors. It supported a quick assessment of potentially affected marbles and helped understand the agent's reasoning. Even though some participants favored the none condition, in general, it was the least helpful due to its lack of explanation, zero transparency, and the introduced increased cognitive effort. Shaking showed some mixed reactions because some found it engaging and attention-grabbing, while others perceived it as distracting and overwhelming for others.



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## Chapter 6

### DISCUSSION

This chapter discusses the findings from the user study results in relation to the previously raised hypotheses. It also highlights their scientific relevance and implications for future research in the field.

#### 6.1 RQ1: Behavioral Trust Calibration

This research question examined whether the presence or type of explanation affects the way users calibrate their trust in the agent’s recommendation. We hypothesized that visual explanation styles will influence users’ trust calibration toward the agent’s recommendation—specifically, increase appropriate trust and reduce overtrust and undertrust.

Based on our quantitative results (Section 5.6), we found that both color and shake explanations led to an increase in appropriate trust and a decrease in undertrust in the agent. This supports hypotheses H1a and H1c. One possible explanation for the increase in appropriate trust is provided by Hoffmann et al. [30]. The authors suggested that when users receive initial instructions about an AI system, they often build an initial mental model of the system. As users interact with the system and get explanations for its actions, they refine their mental model. This can help to better align their reliance on the agent [30].

Regarding undertrust, users rejected the agent’s recommendation even when the agent’s choice of stick was objectively better than the user’s choice. This can occur if there is a lack of sufficient insights into the agent’s behavior. They feel uncertain about the consequences of the recommended action. The game involved complex physics dynamics, and an incorrect stick selection would lead to a cascading effect of marbles falling down. This makes it difficult to choose a stick and predict its outcome without any visual explanation (color or shake). Highlighting marbles with color or shaking motion helped participants to recognize when the agent’s recommendation was correct. Thus, the explanations likely reduced the uncertainty and decreased false rejection of the correct recommendation.

The results did not fully support overtrust hypothesis H1b. Participants followed the agent's recommendation even when it was objectively equal to or worse than their own choice. We expected visual explanations would help to reduce such overreliance on the agent. However, analysis showed that the color explanation increased the overtrust as compared to the no-explanation condition. Whereas the shake explanation did not show a comparable increase or decrease in overtrust as compared to the no-explanation. As the color explanation used a highly conventional and semantic mapping (green = stability/safety and red = instability/risky), it provided a highly intuitive cue and was easy to understand. This may have suggested a stronger sense of confidence in the agent's recommendation and to follow it. Conversely, the shaking explanation primarily highlights marbles that contribute negatively to the agent's decision. This can be perceived as a warning cue, causing users to be more cautious while making a decision, which could be the reason why it does not increase overtrust in the same way.

## 6.2 RQ2: Direction of Trust Change Across Phases

This research question examined whether the presence or type of explanation influenced users to update their subjective trust via the trust meter across the three phases within each game scene. Overall, the explanation styles did not necessarily change trust in the same way at every phase. Instead, it depended on whether users were reacting to the execution of the agent's and users' stick or to the revelation of the optimal solution.

According to the quantitative results, no-explanation showed a larger increase in trust as compared to color and shake when users moved from the before-execution phase to the after-execution phase. This supports hypothesis H2a that the explanation style changes how users update trust after seeing the consequences of the agent's and user's stick. A reasonable explanation is that explanations provide informative cues before the execution phase, allowing participants to initially set a higher value in the trust meter. When the users observed the outcome, it produced less "surprise". This resulted in a smaller trust update. In contrast, participants relied more on their own intuition in the no-explanation condition. They initially set a lower trust value on the trust meter. But once they observed the outcome, it led to a larger trust shift. Thus, explanations can affect trust dynamics by reducing uncertainty and stabilizing reliance, compared to simply increasing trust in the no-explanation condition.

When the user saw the optimal solution, quantitative results revealed that no pairwise comparisons were significant. This does not support hypothesis H2b. One plausible explanation is that after the execution phase, participants may have already reflected on their final trust value by moving the slider of the trust meter close to the value where they expected it to be. Revealing the optimal solution added very little new information to further shift trust. Thus, explanation effects are strongest during earlier phases when they predict outcomes and see the execution phase. Whereas in the later phase, trust update is constrained due to reduced remaining uncertainty.

For the overall trust change across the entire scene, the evidence partially supports hypothesis H2c, suggesting that the explanation conditions can lead to different overall trust trajectories across the entire scene. In particular, no-explanation showed a larger overall increase in trust than in color and shake. A possible reason is that color shake already provides informative visual cues at the early stages in the scene, which stabilizes users expectations and leads to

smaller trust corrections after revealing the optimal solution, resulting in a stronger net increase in baseline.

### 6.3 RQ3: Trust Change in Correlation to Outcome of Agent's Stick

This research question checks whether the magnitude and direction of trust change after execution of the agent's sticks depends on the agent's error severity. We defined this error severity as the outcome of pulling out the agent's recommended stick. There are three severity levels: 0, 1, or 2 marbles dropped. We believe that users may update their trust differently depending on how badly the agent performed. Thus, we hypothesized that the trust change after execution of the agent's recommended stick would differ across different severity levels. Higher severity errors would lead to a stronger decrease in trust.

According to the descriptive pattern in Figure 5.19 and quantitative results, users tend to increase their trust when the agent's recommended stick drops 0 marbles. However, when the agent's stick drops 1 marble, trust change becomes negative, and it decreases further when it drops 2 marbles. This suggests that users do not just check whether the agent was correct or incorrect. But they also update their trust in a graded manner based on the number of marbles dropped. Users used the outcome of the agent's stick as a strong indicator of the agent's competence. So, even if 1 marble is dropped, users perceive it as a failure because it indicates instability and increases the risk of a cascade of additional marbles dropping. Therefore, dropping 2 marbles would mean a stronger trust violation. This also explains the positive trust change when the agent causes no marbles to fall, as the correct outcome would reinforce the users' confidence and the agent is reliable in that case. Thus, error severity directly influences how users update their trust immediately after execution of the agent's stick.

### 6.4 RQ4: Explanation Satisfaction

This research question examined whether the explanation style affected users' satisfaction during gameplay. It checked whether participants felt satisfaction with the agent's explanations while interacting with the game. We hypothesized that user satisfaction would not be equal across the three explanation conditions (none, color, shake); at least one condition would differ from another in overall satisfaction. We believe that user satisfaction should be higher with color and shake explanation as compared to no explanation.

Overall, the quantitative results suggested that both color and shake conditions produced higher mean satisfaction scores than the no-explanation condition. This supports the hypothesis H4. This pattern suggests that the primary gain in satisfaction came from having an explanation. Both visual styles increased the transparency of the agent's reasoning process. Another plausible explanation is that both color and shake explanations satisfied key attributes of ESS. Explanations provided marble-level cues such as color highlights and shaking, which made the agent's recommendation more transparent. It was sufficiently detailed enough to improve understandability and to justify the agent's action. This also improved the overall feeling of satisfaction. Additionally, the explanations were useful because they could directly

compare multiple sticks just by hovering on sticks one by one.

## **6.5 Visual Explanation: Colored Marbles and Shaking Marbles**

Across all three research questions, there was no consistent significant difference between color and shake condition on trust and satisfaction. For trust calibration, both styles increased appropriate trust and reduced undertrust. But they differ in case of overtrust as color increased overtrust and shake showed no significant effect for reducing overtrust. For subjective trust dynamics, both styles showed similar trust update patterns across phases. Furthermore both styles led to similar high satisfaction score.

Overall, this suggests that the primary benefit comes from the presence of explanation rather than from specific visual encoding which was used to present it. Both styles communicates the same underlying information about the agent's reasoning. Therefore, in case of satisfaction, both styles support the same ESS attributes. However, overtrust results indicate that color explanation has strong conventional, semantic meaning and maybe more persuasive. In some cases, it can encourage users to over rely on the agent. In conclusion, both visual explanation styles are generally effective, explanation designs compared to no-explanation.

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# Chapter 7

## SUMMARY

### 7.1 Conclusion

This thesis aimed to advance explainable reinforcement learning from common visual explanations in 2D environments to a 3D, physics-based interactive environment. We used the *Ouch!* Marble Stick game as a case study. We implemented a reinforcement learning agent for the game and embedded two SHAP-based visual explanation styles, such as colored marbles and shaking marbles, directly into the 3D game environment. We investigated whether these explanation styles encouraged users to rely on the agent. Further, we also examined whether the explanations helped to appropriately calibrate their trust in the XRL agent while making decisions alongside the agent. Lastly, we checked whether these explanations improved user satisfaction.

The findings suggest that the effect of visual explanations is strongly linked to the interaction process. In this 3D game setting, users first formed expectations before the execution phase. Then they observed the consequences during the execution phase and revised their confidence in the agent based on the outcome of the agent’s decision. This means explanations are not only defined by what they show but also by when they are presented and how they are delivered. This thesis demonstrated that visual explanations can be meaningfully integrated into a 3D interactive environment and evaluated using user-centric measures.

Trust was measured in two complementary ways during the user study. Subjective trust was measured via trust meters across different phases within a scene. The behavioral trust calibration was based on the outcome of both users’ and the agent’s stick choices and whether users decided to follow or reject the agent’s recommendation. The results indicate that users update their trust most strongly when they witness the consequence of the agent’s decision. In addition, trust changes were also sensitive to the severity of the outcome of the agent’s stick. There was a larger decrease in trust as more number of marbles dropped. This highlights that the outcomes of the agent’s decision play a central role in shaping trust over time. Thus, explanations can shape users’ expectations and reasoning before an action is taken, but the

observed consequences mainly guide how trust is adjusted during interaction.

In terms of behavioral trust calibration, the explanation conditions increased the appropriate trust and reduced undertrust. This indicates that users were more willing to rely on the agent when its recommendations were correct. However, the findings also suggest that the explanations did not reduce overtrust. In particular, the color explanation sometimes encouraged users to over-rely on the agent even when the recommendations were not optimal. This suggests that highly intuitive visual cues can also increase the risk of overreliance on the agent.

The quantitative results also suggested that both color and shake conditions produced higher mean satisfaction scores than the no-explanation condition. However, users mostly preferred the color explanation over shaking explanation, as color cues were clearer and faster to interpret. Thus, both visual explanation styles improved the transparency of the agent and reduced the effort of deciding whether to rely on the agent. This was reflected in both the quantitative results and the participant’s qualitative feedback.

In conclusion, this thesis contributes to research on integrating SHAP-based visual explanations directly into 3D environments and evaluating their effects through a controlled user study. The findings suggest that explainability in 3D interactive settings can improve perceived transparency and user experience.

## 7.2 Limitations and Future Work

While this thesis demonstrates that SHAP-based visual explanations can be embedded into a 3D, physics-based RL game and evaluated using user-centric measures, several limitations should be considered when interpreting the findings. First, the study investigated only a small set of explanation modalities along with a no-explanation baseline: colored marbles and shaking marbles. Both explanation styles encode similar underlying information (SHAP feature contributions) and were designed to provide local explanations. This limits the extent to which we can generalize the conclusions to other explanation formats (e.g., textual rationales, counterfactual explanations, uncertainty displays, or goal/strategy summaries) and to other 3D tasks beyond this game context.

Second, we provided only local explanations that reason about a single stick choice within one scene. However, we did not provide a higher-level understanding of the agent’s overall strategy. Prior XAI work suggests that combining local and global explanations can help users understand both the local decisions of the AI agent, and summarize the overall strategy [4]. Extending the system with global summaries (e.g., a scene-level “strategy sketch”) alongside local cues could improve interpretability and transparency of the agent.

Third, the task complexity is both a strength and a limitation. The physics-based setting created a realistic and observable consequences. However, the perceptual complexity of the marble-stick configuration may also amplify ambiguity in how users interpret explanations. For example, shaking can be experienced as either a helpful attention cue or as distracting noise depending on the participant and the scene, which likely contributed to the mixed qualitative feedback. Future designs could explicitly control complexity (e.g., systematically varying the number of marbles/sticks, stability of initial configurations, or visual occlusion) to determine when explanations remain helpful and when they become overwhelming.



Fourth, the user study sample size and population limit generalizability. Although the Latin square helps to reduce ordering effects in a within-subject study design, the study still involved a relatively small number of participants ( $N=18$ ). A larger and more diverse sample pool would allow stronger claims about general adoption, differences across expertise levels, and robustness of preference patterns.

A promising direction for future work is to explicitly capture and compare human mental models with the agent's explanation. In the current setup, participants choose sticks and provide outcome predictions, but they are not asked to articulate why they chose a stick. Adding short prompts such as "What made you choose this stick?" (or a lightweight tagging interface: "focused on top marbles / center marbles / crowded region / etc.") would enable a direct comparison between user reasoning and SHAP-based attributions. This would make it possible to test whether mental-model alignment helps to calibrate trust: if a participant's reasoning aligns with what the agent highlights, they may be more willing to rely on the recommendation appropriately; if it diverges, they may show undertrust or interpret explanations as misleading.

Similarly, future studies could compare high performers and low performers to examine whether explanation effects depend on participants' underlying strategy and how closely it aligns with the agent. If users' mental model of what makes a stick "safe" matches with the agent's mental model, then high performers may follow a decision strategy that is more similar to the agent's policy. These performers may find the agent's recommendations more predictable and consistent with their own reasoning. This could lead to higher trust and more appropriate reliance on the agent. Low performers, on the other hand, may use less effective strategies, which could lead to weaker alignment with the agent's strategy and potentially lower trust in the agent. Thus, similarities in users and agents's mental models may help high performers to calibrate their trust appropriately.

Overall, the limitations highlight that explainability in 3D reinforcement learning is not only a question of adding visual cues but of designing explanations that remain interpretable under realistic interactions, task complexity, and individual differences in user strategies. To address the limitations outlined above, expand explanation modalities, combine local and global information, increase participant diversity, and explicitly measure mental-model alignment. It would strengthen the generalizability of the findings and provide a deeper insight into how explanations shape calibrated trust in interactive environments. In conclusion, this thesis provides a step toward understanding and evaluating explainable reinforcement learning in 3D settings, showing that visual explanations can support user experience and trust calibration, while also motivating future work toward more robust, transparent, and user-adaptive explanation designs.



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# List of Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>AI</b>   | Artificial Intelligence                         |
| <b>DDPG</b> | Deep Deterministic Policy Gradient              |
| <b>DL</b>   | Deep Learning                                   |
| <b>DQN</b>  | Deep Q-Learning                                 |
| <b>DRL</b>  | Deep Reinforcement Learning                     |
| <b>ESS</b>  | Explanation Satisfaction Scale                  |
| <b>HER</b>  | Hindsight Experience Replay                     |
| <b>LIME</b> | Local-Interpretable Model-agnostic Explanations |
| <b>LRP</b>  | Layer-wise Relevance Propagation                |
| <b>M</b>    | Mean                                            |
| <b>ML</b>   | Machine Learning                                |
| <b>PPO</b>  | Proximal Policy Optimization                    |
| <b>RL</b>   | Reinforcement Learning                          |
| <b>RQ</b>   | Research Question                               |
| <b>SD</b>   | Standard Deviation                              |
| <b>SDRL</b> | Standard Deep Reinforcement Learning            |
| <b>SDT</b>  | Soft Decision Tree                              |
| <b>SHAP</b> | SHapely Additive exPlanations                   |
| <b>XAI</b>  | eXplainable Artificial Intelligence             |
| <b>XRL</b>  | Explainable Reinforcement Learning              |





## A Study Information Sheet

### Introduction

Hi! My name is Mrudula Ankush Ingale, and I'm a student at Saarland University. This study is part of my Master's thesis titled "Evaluating User-Centric Aspects with Visual Explanations in Reinforcement Learning: A Game Case Study."

I invite you to take part in this research. Some terms might sound technical, so please feel free to stop me anytime if something isn't clear — I'll be happy to explain it. You can also ask questions at any point during the study.

### Purpose of the research

The purpose of the study is to evaluate how different 3D visual explanation styles influence user's trust, satisfaction, and understanding when interacting with an Artificial Intelligence (AI) system.

### Participant selection

To participate, you should be 18 years or older and should not have colour blindness, since the study involves colour-based visual elements.

### Participation and Right to Refuse or Withdraw

Your participation in this research is completely voluntary. It is entirely your choice whether you want to participate or not. You can stop or withdraw your consent from the study at any moment without giving any reasons, by notifying the person conducting the study.

### Study Procedure

The research would take approximately 60 minutes.

1. **Introduction:** I will begin by explaining the purpose and structure of the study.
2. **Consent:** You will receive a consent form to read and sign before we begin.
3. **Game Instructions:** I will explain the rules of the 3D game and how it works.
4. **Study Conditions:** There are three parts (study conditions) in this study. At the start of each one, I will show you an example to help you understand how to play and how to interpret the visual explanations.

5. **Questions:** Before you start the main part of the game in each condition, you'll have a chance to ask any questions, and I'll answer them.
6. **Game Play:** You will then play the game under the different study conditions.
7. **Questionnaires:** After finishing each condition, you will fill out a short questionnaire about your trust, understanding, and satisfaction with the explanations provided by the AI assistant.

## **Risks**

Your participation in this study should not entail any significant risks. The only potential discomfort may be mild mental fatigue from playing the game. To mitigate these risks, you will be allowed to take regular breaks to make sure you feel comfortable throughout the study.

## **Reimbursements**

You will not receive any monetary compensation for participating in this research. However, as a token of appreciation for your valuable time, snacks and sweets will be provided.

## **Confidentiality of the data**

With your consent, the following data will be collected and processed anonymously:

- Basic demographic data (e.g. gender, age, previous Explainable AI experiences, etc.)
- Answers to questions about the user experience you will be asked during the study as well as further measurements and results of the study. These questions cover how you experienced the interaction with the Explainable AI system.

All information you provide will be made anonymous and stored securely in password-protected systems. Only the research team will have access to it. The data will be kept until the end of my thesis after the completion of the study; all personal data will be securely deleted following GDPR-compliant practices.

## **Sharing the Results**

The findings from this study will be included in my master's thesis and may also be used to inform future research. Your personal identity will never be revealed, and no confidential information will be shared.

## **Who to Contact**

If you have any questions about the study at a later stage, feel free to contact:

- Mrudula Ingale (mrin00001@stud.uni-saarland.de)
- Advisor: Gian-Luca Kiefer (gian-luca.kiefer@dfki.de)

If you have any questions that are not answered by this data protection declaration, or if you would like further information on data protection, please contact our data protection officer (datenschutz@dfki.de).

## **B Certificate of Consent**

### **Statement by the participant**

I have read and understood the information about the study. I have had the opportunity to ask questions about it. I have received answers to all my questions about the study. I consent voluntarily to participate in this study.

**Participant ID:** \_\_\_\_\_

**Signature of Participant:** \_\_\_\_\_

**Date:** \_\_\_\_\_  
*Day/month/year*

### **Statement by the researcher/person taking consent**

I confirm that the participant was given an opportunity to ask questions about the study, and all the questions asked by the participant have been answered correctly and to the best of my ability. I confirm that the individual has not been coerced into giving consent, and the consent has been given freely and voluntarily.

**Name of person taking the consent:** Mrudula Ankush Ingale

**Signature of person taking the consent:** \_\_\_\_\_

**Date:** \_\_\_\_\_  
*Day/month/year*

## C Pre-Study Questionnaire

Please fill in the following information. All responses are anonymous and used only for research purposes.

### Basic Information

Participant ID (assigned by researcher): \_\_\_\_\_

Age: \_\_\_\_\_ years

### Gender:

- ☐ Male
- ☐ Female
- ☐ Non-binary / Other
- ☐ Prefer not to say

### What is the highest level of education you have completed?

- ☐ Upper secondary education (higher secondary/junior college, Std. 11–12, HSC/PUC)
- ☐ Post-secondary non-tertiary education (polytechnic/ITI diploma, vocational training after Std. 10 or 12)
- ☐ Short-cycle tertiary education (associate-level diploma or equivalent, rare in India)
- ☐ Bachelor's degree
- ☐ Master's degree or higher
- ☐ Doctoral degree
- ☐ Other: \_\_\_\_\_

### Game & Technology Background

#### How often do you play video or computer games?

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Often
- ☐ Very often

Have you played puzzle or physics-based games (e.g., Jenga, Tumbling Monkey, KerPlunk, etc.) before?

☐ Yes

☐ No

**Do you have any form of color vision deficiency (e.g., color blindness)?**

☐ Yes

☐ No

### **AI & Explanation Familiarity**

**Do you have experience with Artificial Intelligence (AI)?**

☐ Yes

☐ No

☐ Maybe

**How familiar are you with Artificial Intelligence (AI)?**

☐ Not at all

☐ Slightly familiar

☐ Moderately familiar

☐ Very familiar

☐ Expert level

**How familiar are you with Explainable Artificial Intelligence (XAI)?**

☐ Not at all

☐ Slightly familiar

☐ Moderately familiar

☐ Very familiar

☐ Expert level

## **C.1 Pre-study questionnaire summary plots**

The following figures summarize participants' responses to the pre-study questionnaire (n=18). These plots are provided for transparency and are not analyzed further in the main chapters.

### Pre-Study Demographics Overview (n=18)

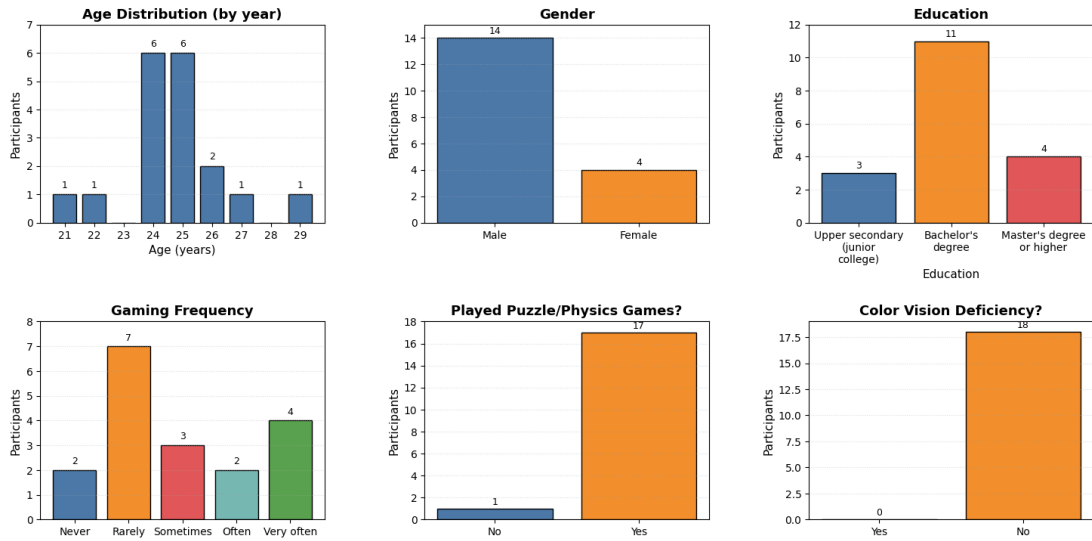


Figure C.1: Pre-study demographics overview (n=18): age distribution, gender, education, gaming frequency, prior experience with puzzle/physics games, and self-reported color vision deficiency

### Familiarity Levels (n=18)

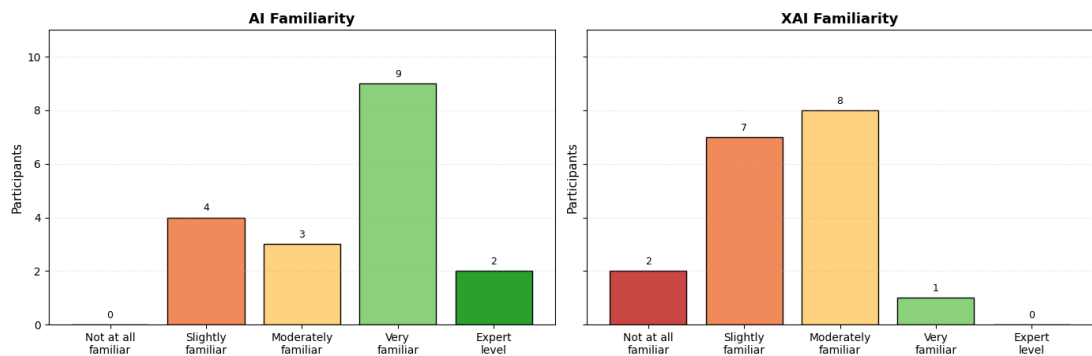


Figure C.2: Self-reported familiarity levels (n=18) with AI and explainable AI (XAI)

## D Explanation Satisfaction Questionnaire (AI Agent)

This questionnaire was adopted from the Explanation Satisfaction Scale (ESS) proposed by Hoffman et al. [30].

Participant ID: \_\_\_\_\_

Study Condition: \_\_\_\_\_

**Instructions:** Please indicate how much you agree or disagree with each of the following statements about the explanation provided by the AI agent in this round.

Use the scale below to rate each item:

1 = Strongly Disagree    2 = Disagree    3 = Neutral    4 = Agree    5 = Strongly Agree

| # | Statement                                                               | 1                        | 2                        | 3                        | 4                        | 5                        |
|---|-------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1 | From the explanation, I know how the AI agent works.                    | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 2 | This explanation of how the AI agent works is satisfying.               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 3 | This explanation of how the AI agent works has sufficient detail.       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 4 | This explanation of how the AI agent works seems complete.              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 5 | This explanation of how the AI agent works tells me how to use it.      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 6 | This explanation of how the AI agent works is useful to my goals.       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 7 | This explanation of the AI agent shows me how accurate the AI agent is. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## E Post-Condition Questionnaire (Open-Ended Questions)

Participant ID: \_\_\_\_\_

Study Condition: \_\_\_\_\_

Please answer the following questions based on your experience in this round.

**1. What aspects of the AI agent's behavior or explanation did you find helpful or clear?**

Write your answer:

---

---

**2. Was there anything confusing, frustrating, or unclear during this round?**

Write your answer:

---

---

**3. Did the explanation (or lack of explanation) help you decide whether to trust or follow the agent's choice? Why or why not?**

Write your answer:

---

---

**4. (Optional) Any suggestions or comments about this round?**

Write your answer:

---

---



## F Post-Study Questionnaire

Participant ID: \_\_\_\_\_

### 1. Explanation Style Preference

Please rank the following explanation styles from **1 (Most Preferred)** to **3 (Least Preferred)** based on your overall experience.

| Explanation Style           | Rank (1-3) |
|-----------------------------|------------|
| No Explanation (Baseline)   | _____      |
| Colored Marbles             | _____      |
| Shaking Marbles (Animation) | _____      |

### 2. Please explain your ranking

Why did you prefer the style you ranked first?

---

---

What made the least preferred style less helpful or enjoyable?

---

---

### 3. Understanding the Agent

In which condition did you best understand why the agent made its decisions?

---

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Please explain what helped you in that condition.

---

---

#### **4. Trust in the Agent**

**In which condition did you trust the agent's decisions the most?**

---

---

**Was there any condition where you found it hard to trust the agent? Please explain.**

---

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#### **5. Usefulness of Explanations**

**Compare the two explanation styles you saw (Color highlights vs. Jitter).**

**Which explanation was more useful or clear to you? Please elaborate (what you liked or disliked about each).**

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#### **6. Additional Comments (Optional)**

**Any other thoughts, feedback, or suggestions about the explanations, the agent, or the study in general?**

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## **G Declaration of AI Tool Usage**

Generative AI tools were used as supportive instruments during the preparation of this thesis. In particular, ChatGPT (GPT-4o and GPT-5) was used:

- To get guidance for developing the game objects and configuring appropriate properties for these objects in Unity and Blender.
- To support debugging and troubleshooting of implementation issues.
- To clarify technical concepts related to explainability methods, Unity ML-Agents Toolkit and the corresponding Python API.
- For refining user study materials and questionnaire wording for clarity and consistency.
- For clarifying the theoretical statistical concepts for the user study analysis.
- To improve the formatting of results and figures.

Additionally, Grammarly and ChatGPT were used for grammar and spelling correction and general minor language polishing.

Consensus.ai was used to suggest possibly relevant research papers during the literature review phase.